

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\183-C.pvdb

Date of Analysis : Sep 9, 2026 9:46am

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User
 FileName : 183-C Page 2 of 163
 Vessel Design Summary: Step: 29 9:46am Sep 9,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1285.000 mm ID
 Vessel Design Length, Tangent to Tangent 8339.15 mm
 Specified Datum Line Distance 0.00 mm
 Shell Side Design Temperature 275 °C
 Channel Side Design Temperature 430 °C
 Shell Side Design Pressure 4350.000 KPa.
 Channel Side Design Pressure 5900.000 KPa.
 Shop Shell Side Test Pressure 6845.276 KPa.
 Field Shell Side Test Pressure 5655.000 KPa.
 Shop Channel Side Test Pressure 8912.892 KPa.
 Field Channel Side Test Pressure 7721.800 KPa.
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	
Head	SA-387 11	2	...	K11789	
Flange	SA-336 F11	3	...	K11572	
Nozzle	SA-336 F11	3	...	K11572	
Nozzle Flg	SA-336 F11	2	...	K11572	
Tubes	SA-213 T11	...	...	K11597	
Tubesheet	SA-336 F11	3	...	K11572	
Flg Bolting	SA-193 B16	...	64 < t <= 100	K14072	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-387 11	2	...	K11789	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Blind Flang	5917.104	103.42	6091.000	3.0000	N/A	No
LHS Ch Flg	5912.596	103.42	6849.000	3.0000	N/A	No
LHS Ch Shell	5912.625	103.42	6849.000	3.0000	N/A	No
Main Shell-1	4362.625	0.00	4710.000	3.0000	N/A	No
Dish End	4362.625	0.00	4587.000	3.0000	No	No

Liquid Level: 1745.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
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Vessel Design Summary:

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Sep 9, 2026

Body Flg	203.3	197.0	205.0	197.0	193.2	...	1.00	1.00
Body Flg	503.4	300.0	248.0	240.0	235.3	...	1.00	1.00
Cylinder	1503.4	1000.0	34.0	34.0	29.6	...	1.00	1.00
Cylinder	8288.3	6600.0	24.0	24.0	22.4	...	1.00	1.00
Ellipse	8339.1	50.8	28.0	23.0	22.0	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	197	165.989	103.419	No Calc
20	30	240	235.255	103.419	No Calc
30	Tsht	34	6.86159	103.419	3373.94
Tsht	40	34	No Calc	103.419	No Calc
40	70	24	No Calc	...	888.024
70	80	23	No Calc	...	1296.7

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	999.97	27264.9	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	70	6757.91	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube Inlet	WRC-297	0.850	Shell	Passed
N2-Tube Outlet	WRC-297	0.857	Shell	Passed
N3-Shell Inlet	WRC-297	0.912	Shell	Passed
N4-Shell Outle	WRC-297	0.919	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	375.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1220.000 mm
Wear Pad Width	575.000 mm
Wear Pad Thickness	14.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1320.000 mm

Baseplate Length	1400.000 mm
Baseplate Thickness	22.000 mm
Baseplate Width	400.000 mm
Number of Ribs (including outside ribs)	4
Rib Thickness	10.000 mm
Web Thickness	10.000 mm
Height of Center Web	560.000 mm
Number of Bolts in Baseplate	6



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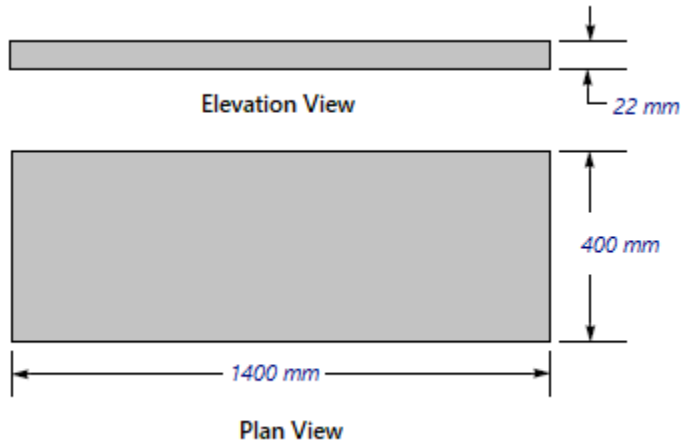
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Vessel Design Summary:

Step: 29

9:46am

Sep 9, 2026

Baseplate Sketch**Summary of Maximum Saddle Loads, Operating Case, Un-Factored:**

Maximum Vertical Saddle Load	32220.18	Kgf
Maximum Transverse Saddle Shear Load	5227.32	Kgf
Maximum Longitudinal Saddle Shear Load	1899.51	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	28018.80	Kgf
Maximum Transverse Saddle Shear Load	167.43	Kgf
Maximum Longitudinal Saddle Shear Load	129.58	Kgf

Weights:

Fabricated - Bare w/o removable internals	20320.9	kg.
Shop Test - Fabricated + water (full)	31361.7	kg.
Shipping - Fab. + rem. intls.+ shipping app.	20320.9	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	21220.9	kg.
Empty - Fab. + intls. + details + wghts.	22857.4	kg.
Operating - Empty + operating liquid (no ca)	32998.2	kg.
Field Test - Empty weight + test medium (full)	32298.8	kg.

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Nozzle Summary:

Step: 27 9:46am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-Tube Inlet	6611	OK	...	OK	8.26	OK	Passed
N2-Tube Outlet	6849	OK	...	OK	7.80	OK	Passed
Vent	6849	...	...	OK	6.42	OK	No Calc[*]
N3-Shell Inlet	4710	...	...	OK	11.33	OK	Passed
N4-Shell Outlet	4710	...	...	OK	11.33	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 4710 Nozzle : N3-Shell Inlet [Shellside]
 Minimum MAWP Nozzles : 6611 Nozzle : N1-Tube Inlet [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N1-Tube Inlet	1203.350	0.000	196.099
30	N2-Tube Outlet	1003.350	180.000	150.900
30	Vent	803.350	0.000	122.325
40	N3-Shell Inlet	5038.350	0.000	612.050
40	N4-Shell Outlet	5038.350	180.000	739.151

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 26 9:46am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
Vent	50 mm	160	WNF	60.3	8.738	...	...	286	...
N2-Tube Outlet	80 mm	160	WNF	88.9	11.125	...	...	288	...
N1-Tube Inlet	100 mm	120	WNF	114.3	11.125	...	...	289	...
N3-Shell Inlet	350 mm	120	WNF	355.6	27.788	...	...	324	...
N4-Shell Outlet	400 mm	80	WNF	406.4	21.412	...	...	339	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Vent	SA-336 F11	30.000	10.000	...	...	...
N2-Tube Out	SA-336 F11	30.000	10.000	...	...	...
N1-Tube Inl	SA-336 F11	30.000	10.000	...	...	...
N3-Shell In	SA-336 F11	24.000	10.000	...	...	...
N4-Shell Ou	SA-336 F11	24.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Vent	803.350	0.0	250.00	0.00	LHS Ch Shell
N2-Tube Outlet	1003.350	180.0	250.00	0.00	LHS Ch Shell
N1-Tube Inlet	1203.350	0.0	250.00	0.00	LHS Ch Shell
N3-Shell Inlet	5038.350	0.0	250.00	0.00	Main Shell-1
N4-Shell Outlet	5038.349	180.0	250.00	0.00	Main Shell-1

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Bill of Material:

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 205.0mm THK X 1745.0mm OD	SA-336 F11
1	WELD NECK FLANGE: 248.0mm THK X 1745.0mm OD	SA-336 F11
1	CYLINDER: 34.0mm THK X 1285.0mm ID X 1000.0mm	SA-387 11
1	ELLIPTICAL HEAD: 2.0 X 1, 28.0mm THK X 1285.0mm ID X 50.8mm	SA-387 11
1	INSULATION: 200mm X 150mm THK	Hot Insulation
1	INSULATION: 280mm X 150mm THK	Hot Insulation
1	INSULATION: 1000mm X 150mm THK	Hot Insulation
10	BAFFLES: 1315.0mm DIA X 8.0mm THK	SA-516 70
8	TIE RODS: 14mm DIA X 4866mm	SA-193 B7M
1	TUBESHEET: 1353.0mm X 185.0mm THK	SA-336 F11
626	TUBES: 6000.0mm X 19.0mm DIA X 2.1mm THK	SA-213 T11
2	SADDLE: 375.0mm X 120 DEG	SA-387 11
1	INSULATION: 6600mm X 100mm THK	Hot Insulation
1	INSULATION: 50mm X 100mm THK	Hot Insulation
1	GASKET: 1359.0mm OD X 1317.0mm ID	...
24	BODY FLANGE BOLTS: 76.2mm DIA	SA-193 B16
48	NUTS FOR BODY FLANGE BOLTS: 76.2mm DIA	...
1	CYLINDRICAL SEGMENT 24.0mm THK X 2576.7mm X 6600.0mm	SA-387 11
1	CYLINDRICAL SEGMENT 24.0mm THK X 1460.2mm X 6600.0mm	SA-387 11
1	NAMEPLATE	...



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MDMT Summary:

Step: 28 9:46am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Main Shell-1 [8]	B	-2	-4		0.971	24.000	1.00	Yes
Dish End [10]	A	16	16		0.997	23.000	1.00	Yes
Dish End [7]	A	22	12		0.815	28.000	1.00	Yes
Left Saddle f[24]	B	-18	-20		0.971	14.000	1.00	Yes
RH saddle fix[24]	B	-18	-20		0.971	14.000	1.00	Yes
N3-Shell Inlet[1]	B	-2	-7		0.920	24.000	1.00	Yes
N4-Shell Outle[1]	B	-2	-6		0.922	24.000	1.00	Yes
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No
Tubesheet: SS[13]	B	15	-5		0.633	46.250	1.00	No

Warmest MDMT: 22 16

LHS Blind Fla[11]	B	16	9		0.876	49.250	1.00	No
LHS Ch Flg [11]	B	7	-4		0.802	34.000	1.00	No
LHS Ch Shell [8]	B	7	1		0.879	34.000	1.00	Yes
N1-Tube Inlet [1]	B	7	-1		0.850	34.000	1.00	Yes
N2-Tube Outlet[1]	B	7	-1		0.852	34.000	1.00	Yes
Vent [1]	B	7	-1		0.850	34.000	1.00	Yes
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No
Tubesheet: CS[14]	B	15	7		0.859	46.250	1.00	No
Bolting [21]		-29						

Warmest MDMT: 16 9

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	16.0	-12.0	Fail
Channel/Tube	9.0	-12.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.



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MDMT Summary:

Step: 28 9:46am Sep 9, 2026

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 9:46am Sep 9, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 9:46am Sep 9,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	4350	KPa.
Channel Side Design Pressure	5900	KPa.
Shell Side Design Temperature	275.0	°C
Channel Side Design Temperature	430.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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Input Echo:

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Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	9500 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Input Echo:

Step: 1 9:46am Sep 9, 2026

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No
Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
Use Table G instead of exact equation for "A" Yes
Shell Head Joints are Tapered Yes
Compute "K" in corroded condition Yes
Use Code Case 2286 No
Use the MAWP to compute the MDMT Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes
Calculate B31.3 type stress for Nozzles with Loads Yes
Reduce the MDMT due to lower membrane stress Yes
Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10
Element To Node	20
Element Type	Flange
Description	LHS Blind Flang
Distance "FROM" to "TO"	197 mm
Flange Outside Diameter	1745 mm
Element Thickness	197 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	205 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5900 KPa.
Design Temperature Internal Pressure	430 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11
Allowable Stress, Ambient	148 N./mm ²
Allowable Stress, Operating	147 N./mm ²
Allowable Stress, Hydrotest	278.99 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	16.002 mm
Yield Stress, Operating	223 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-3
UNS Number	K11572
Class / Thickness / Grade	3 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1745 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	10
Detail Type	Insulation
Detail ID	Insulation



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Input Echo:

Step: 1 9:46am Sep 9,2026

Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	200	mm
Thickness of Insulation	150	mm
Density	0.000352	kg./cm ³

Element#	2/5
Element From Node	20
Element To Node	30
Element Type	Flange
Description	LHS Ch Flg
Distance "FROM" to "TO"	300 mm
Flange Inside Diameter	1285 mm
Element Thickness	240 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	248 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5900 KPa.
Design Temperature Internal Pressure	430 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1285 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	280 mm
Thickness of Insulation	150 mm
Density	0.000352 kg./cm ³

Element#	3/5
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	LHS Ch Shell
Distance "FROM" to "TO"	1000 mm
Inside Diameter	1285 mm
Element Thickness	34 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	34 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	5900 KPa.
Design Temperature Internal Pressure	430 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Allowable Stress, Ambient	148 N./mm ²
Allowable Stress, Operating	147 N./mm ²
Allowable Stress, Hydrotest	279 N./mm ²
Material Density	0.00775 kg./cm ³



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Input Echo:

Step: 1 9:46am Sep 9,2026

P Number Thickness	16.002	mm
Yield Stress, Operating	223	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1285	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1000	mm
Thickness of Insulation	150	mm
Density	0.000352	kg./cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N1-Tube Inlet	
Dist. from "FROM" Node / Offset dist	700	mm
Nozzle Diameter	100	mm
Nozzle Schedule	120	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	33.297	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N2-Tube Outlet	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	80	mm
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	23.805	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	Vent	
Dist. from "FROM" Node / Offset dist	300	mm
Nozzle Diameter	50	mm
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	



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Input Echo:

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Element From Node	30
Detail Type	Weight
Detail ID	Pass partition we
Dist. from "FROM" Node / Offset dist	500 mm
Miscellaneous Weight	252 Kg
Offset from Element Centerline	0 mm

Element#	4/5
Element From Node	40
Element To Node	70
Element Type	Cylinder
Description	Main Shell-1
Distance "FROM" to "TO"	6600 mm
Inside Diameter	1285 mm
Element Thickness	24 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4350 KPa.
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	275 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	Left Saddle flo
Dist. from "FROM" Node / Offset dist	1320 mm
Width of Saddle	375 mm
Height of Saddle at Bottom	1220 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	575 mm
Thickness of Wear Plate	14 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.30000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	40
Detail Type	Saddle
Detail ID	RH saddle fixed
Dist. from "FROM" Node / Offset dist	5280 mm
Width of Saddle	375 mm
Height of Saddle at Bottom	1220 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	575 mm
Thickness of Wear Plate	14 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm



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Input Echo:

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Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1285 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	40
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	6600 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	40
Detail Type	Nozzle
Detail ID	N3-Shell Inlet
Dist. from "FROM" Node / Offset dist	3350 mm
Nozzle Diameter	350 mm
Nozzle Schedule	120
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	233.86 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11

Element From Node	40
Detail Type	Nozzle
Detail ID	N4-Shell Outlet
Dist. from "FROM" Node / Offset dist	3350 mm
Nozzle Diameter	400 mm
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	267.87 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11

Element From Node	40
Detail Type	Weight
Detail ID	Fireproofing
Dist. from "FROM" Node / Offset dist	1320 mm
Miscellaneous Weight	450 Kg
Offset from Element Centerline	1200 mm

Element From Node	40
Detail Type	Weight
Detail ID	Fireproofing we
Dist. from "FROM" Node / Offset dist	5280 mm
Miscellaneous Weight	450 Kg
Offset from Element Centerline	1200 mm

Element#	5/5
Element From Node	70
Element To Node	80
Element Type	Elliptical
Description	Dish End
Distance "FROM" to "TO"	50.8 mm
Inside Diameter	1285 mm



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Input Echo:

Step: 1 9:46am Sep 9,2026

Element Thickness	23	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	28	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4350	KPa.
Design Temperature Internal Pressure	275	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	275	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1285	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	70	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	50.5	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

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XY Coordinate Calculations:

Step: 2 9:46am

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Blind Flang		203.35	...	197	...
LHS Ch Flg		503.35	...	300	...
LHS Ch Shell		1503.35	...	1000	...
Main Shell-1		8288.35	...	6600	...
Dish End		8339.15	...	50.8	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: New Flange

Flng: 5 9:46am Sep 9,2026

Flange Input Data Values

Description: New Flange :

LHS Blind Flang

Flange Type	TEMA Channel Cover		
Design Pressure	P	5917.10	KPa.
Design Temperature		430	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1745.000	mm
Flange Thickness	t	197.0000	mm
Flange Material		SA-336 F11	
Flange Material UNS number		K11572	
Flange Allowable Stress At Temperature	Sfo	147.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00	N./mm ²
Bolt Material		SA-193 B16	
Bolt Allowable Stress At Temperature	Sb	151.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	152.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Perimeter along the Center of the Bolts	L	4979.425	mm
Youngs Modulus for Blind Flange Matl.	Sy	176184.59	N./mm ²
Allowed Channel Cover Deflection	dfAllow	1.6062	mm
Diameter of Bolt Circle	C	1585.000	mm
Nominal Bolt Diameter	a	76.2000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	1370.000	mm
Flange Face Inside Diameter	Fid	1285.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1359.000	mm
Gasket Inside Diameter	Gi	1317.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1370.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



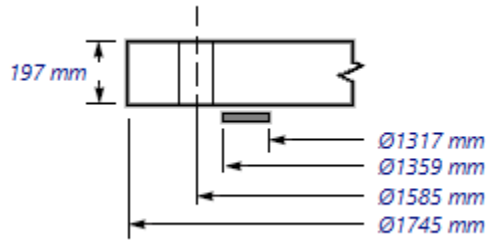
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Flg Calc [Int P]: New Flange

Flng: 5 9:46am Sep 9,2026



ASME Code, Section VIII Division 1, 2025

Gasket Contact Width, $N = (G_o - G_i)/2$ 21.000 mm
 Basic Gasket Width, $b_o = N/2$ 10.500 mm
 Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$ 8.165 mm
 Gasket Reaction Diameter, $G = G_o - 2 \cdot b$ 1342.669 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 1342.6689^2 \cdot 5917.104$
 $= 854345.688 \text{ Kg}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:
 $= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$
 $= 2 \cdot 8.1655 \cdot \pi \cdot 1342.6689 \cdot 3.0 \cdot 5917.1$
 $+ 2 \cdot 1370.0 \cdot 5.0 \cdot 3.0 \cdot 5917.1045$
 $= 149497.094 \text{ Kg}$

Design Bolt Load for the Operating Condition [Wo]:
 $= \max(W_o,1 + W_o,2 + H'p, 0)$
 $= \max(854346 + 149497 + 0, 0)$
 $= 1003842.750 \text{ Kg}$

Design Bolt Load for the Gasket Seating Condition [Wg]:
 $= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$
 $= 68.95 \cdot 8.1655 \cdot \pi \cdot 1342.669 + 68.95 \cdot 5.0 \cdot 1370.0$
 $= 290322.906 \text{ Kg}$

Required Bolt Area [Am]:
 $= \max((W_o + F_a + 4Me/G)/S_b, W_g/S_a)$
 $= \max((1003843 + 0 + 4 \cdot 0/1342.67)/151, 290323/152)$
 $= 651.936 \text{ cm}^2$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	651.936	979.198	
Radial Distance between Bolts and Edge:	73.025	80.000	
Circ. Spacing between the Bolts:	158.750	206.884	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
 $= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$
 $= 979.198 \cdot 152.0 / (68.95 \cdot \pi (1359.0 + 1317.0))$
 $= 25.677 \text{ mm}$ [Note: Exceeds actual gasket width, 21.000]

Flange Design Bolt Load, Gasket Seating [W or Wgs]:
 $= S_a \cdot A_b$
 $= 152.0 \cdot 979.1981$



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Flg Calc [Int P]: New Flange

Flng: 5 9:46am

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= 1517670.88 Kgf

Gasket Load for the Operating Condition [HG]:

= W_o

= 1003842.75 Kgf

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

= (C - G) / 2

= (1585.0 - 1342.6689) / 2

= 121.1655 mm

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

= $1.9 \cdot W_{m1} \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$

= $1.9 \cdot 1003843 \cdot 121.1655 \cdot 1.0 / (194.0^2 \cdot 1342.6689) + 0.3 \cdot 1.0 \cdot 5917.1 \cdot 1342.6689^2 / 194.0^2$

= 129.88 N./mm²

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

= $1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$

= $1.9 \cdot 1517671 \cdot 121.1655 \cdot 1.0 / (194.0^2 \cdot 1342.6689)$

= 67.81 N./mm²

Bolt Stress, Operating [BS_o]:

= W_{m1} / A_b

= 1003843 / 979.1981

= 100.54 N./mm²

Bolt Stress, Seating [BS_a]:

= (W_{m2} / A_b)

= (290323 / 979.1981)

= 29.08 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	129.88	147.00	67.81	148.00
Bolting	100.54	151.00	29.08	152.00

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 1.5853 mm

Allowed deflection at center of Cover 1.6062 mm

Reqd. Blind Flange Thickness at Center 185.353 mm

Reqd. Blind Flange Thickness at Gasket 134.313 mm

Min. Required Flange Thickness per TEMA 193.153 mm

Estimated M.A.W.P. (Operating) 6108 KPa.

Estimated Finished Weight of Flange at given Thk. 3651.5 kg.

Estimated Unfinished Weight of Forging at given Thk 3651.5 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.876, Temperature Reduction per Fig. UCS 66.1 = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 16 °C

Min Metal Temp. at Required thickness (UCS 66.1) 9 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:



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Flg Calc [Int P]: New Flange

Flng: 5

9:46am

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Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Tangential Flange	125.96	148.00	65.76	148.00
Bolting	100.54	152.00	29.08	152.00

Minimum Req'd Thickness, ambient, no Corrosion: 184.175 mm

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Flg Calc [Int P]: New Flange

Flng: 6 9:46am Sep 9,2026

Flange Input Data Values

Description: New Flange :

LHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	5912.60	KPa.
Design Temperature		430	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1285.000	mm
Flange Outside Diameter	A	1745.000	mm
Flange Thickness	t	240.0000	mm
Thickness of Hub at Small End	go	34.0000	mm
Thickness of Hub at Large End	gl	51.0000	mm
Length of Hub	h	60.0000	mm
Flange Material	SA-336 F11		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	147.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00	N./mm ²
Bolt Material	SA-193 B16		
Bolt Allowable Stress At Temperature	Sb	151.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	152.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	1585.000	mm
Nominal Bolt Diameter	a	76.2000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	1370.000	mm
Flange Face Inside Diameter	Fid	1285.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1359.000	mm
Gasket Inside Diameter	Gi	1317.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1317.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



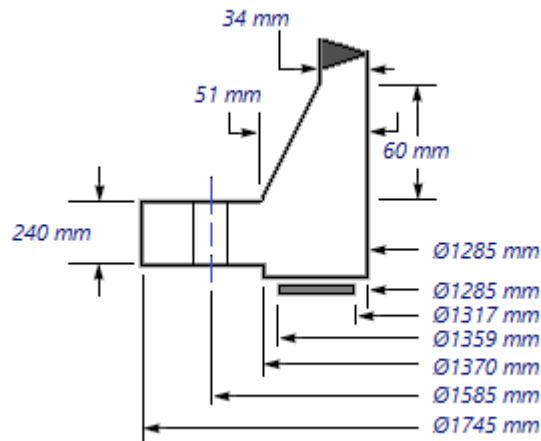
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Flg Calc [Int P]: New Flange

Flng: 6 9:46am Sep 9, 2026



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (5912.6 \cdot (1285.0/2 + 3.0)) / (147.0 \cdot 1.0 - 0.6 \cdot 5912.6) + Ca$$

$$= 29.6068 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (147.0 \cdot 1.0 \cdot 31.0) / (645.5 + 0.6 \cdot 31.0)$$

$$= 6861.520 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (148.0 \cdot 1.0 \cdot 34.0) / (642.5 + 0.6 \cdot 34.0)$$

$$= 7590.448 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	237.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1291.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	48.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	31.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	99.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	21.000 mm
Basic Gasket Width, $b_o = N/2$	10.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.165 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1342.669 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1342.6689^2 \cdot 5912.596$$

$$= 853694.688 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 8.1655 \cdot \pi \cdot 1342.6689 \cdot 3.0 \cdot 5912.6$$

$$+ 2 \cdot 1317.0 \cdot 5.0 \cdot 3.0 \cdot 5912.5957$$

$$= 148424.500 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1291.0^2 \cdot 5912.5957$$

$$= 789254.688 \text{ Kg}$$



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Flg Calc [Int P]: New Flange

Flng: 6 9:46am

Sep 9, 2026

Pressure Force on Flange Face [Ht]:

$$\begin{aligned}
 &= H - H_d \\
 &= 853695 - 789255 \\
 &= 64440.004 \text{ Kgf}
 \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned}
 &= \max(W_{o,1} + W_{o,2} + H'p, 0) \\
 &= \max(853695 + 148425 + 0, 0) \\
 &= 1002119.188 \text{ Kgf}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 68.95 \cdot 8.1655 \cdot \pi \cdot 1342.669 + 68.95 \cdot 5.0 \cdot 1317.0 \\
 &= 288459.750 \text{ Kgf}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4M_e/G) / S_b, W_g / S_a) \\
 &= \max((1002119 + 0 + 4 \cdot 0 / 1342.67) / 151, 288460 / 152) \\
 &= 650.816 \text{ cm}^2
 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$\begin{aligned}
 &= 2a + 6t / (m + 0.5) \\
 &= 2 \cdot 76.2 + 6 \cdot 237.0 / (3.0 + 0.5) \\
 &= 558.686 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 1585.0 \cdot \sin(\pi / 24) \\
 &= 206.884 \text{ mm}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{ (B_s / (2a + t)) }, 1) \\
 &= \max(\sqrt{ (206.884 / (2 \cdot 76.2 + 237.0)) }, 1) \\
 &= 1.0000
 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	650.816	979.198	
Radial Distance between Hub and Bolts:	92.075	99.000	
Radial Distance between Bolts and Edge:	73.025	80.000	
Circ. Spacing between the Bolts:	158.750	206.884	558.686

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 979.198 \cdot 152.0 / (68.95 \cdot \pi (1359.0 + 1317.0)) \\
 &= 25.677 \text{ mm} [\text{Note: Exceeds actual gasket width, 21.000}]
 \end{aligned}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a \cdot A_b \\
 &= 152.0 \cdot 979.1981 \\
 &= 1517670.88 \text{ Kgf}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o - H \\
 &= 1002119 - 853695 \\
 &= 148424.50 \text{ Kgf}
 \end{aligned}$$



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Flg Calc [Int P]: New Flange

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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1585.0 - 1342.6689) / 2$$

$$= 121.1655 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((1585.0 - 1291.0) / 2 + 121.1655)$$

$$= 134.0827 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1585.0 - 1291.0 - 48.0) / 2$$

$$= 123.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	789255.	123.0000	1.0000	97080.
Face Pressure, Mt	64440.	134.0827	1.0000	8640.
Gasket Load, Mg	148424.	121.1655	1.0000	17984.
Gasket Seating, Matm	1517671.	121.1655	1.0000	183893.

Total Moment for Operation, Mo 123705. Kg-m.

Total Moment for Gasket seating, Mg 183893. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 200.053 mm

Hub Ratio, h/h0 = HL / H0 0.300

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.548

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.880

Factor V per 4.16.5 0.361

Factor f per 4.16.5 1.249

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1745.0 / 1291.0$$

$$= 1.352$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.352^2 \cdot \log_{10}(1.352))) / (1.352^2 - 1)) / (1.352 - 1)$$

$$= 6.601$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.352^2 (1 + 8.55246 \cdot \log_{10}(1.352)) - 1) / ((1.0472 + 1.9448 \cdot 1.352^2) \cdot (1.352 - 1))$$

$$= 1.775$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.352^2 (1 + 8.55246 \cdot \log_{10}(1.352)) - 1) / (1.36136 (1.352^2 - 1) \cdot (1.352 - 1))$$

$$= 7.254$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.352^2 + 1) / (1.352^2 - 1)$$

$$= 3.418$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$



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Flg Calc [Int P]: New Flange

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$$= 7.254 \cdot 31.0^2 \cdot 200.053 / 0.361$$

$$= 3857819.500 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.88 / 200.053$$

$$= 0.004 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (237.0 \cdot 0.004 + 1) / 1.775 + 237.0^3 / 3857820$$

$$= 4.601$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot M_{op} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1.2493 \cdot 123705 / 1291.0) / (4.6013 \cdot 48.0^2)$$

$$= 110.73 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot M_{atm} / B_{cor}) / (L \cdot g_1^2)$$

$$= (1.2493 \cdot 183893 / 1291.0) / (4.6013 \cdot 48.0^2)$$

$$= 164.61 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot M_{op} / B_{cor}) / (L \cdot t^2)$$

$$= (2.39 \cdot 123705 / 1291.0) / (4.6013 \cdot 237.0^2)$$

$$= 8.69 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot M_{atm} / B_{cor}) / (L \cdot t^2)$$

$$= (2.39 \cdot 183893 / 1291.0) / (4.6013 \cdot 237.0^2)$$

$$= 12.92 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot M_{op} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ro}$$

$$= (6.6009 \cdot 123705 / (237.0^2 \cdot 1291.0)) - 3.4184 \cdot 9$$

$$= 80.72 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot M_{atm} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ra}$$

$$= (6.6009 \cdot 183893 / (237.0^2 \cdot 1291.0)) - 3.4184 \cdot 13$$

$$= 120.00 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (111 + \max(9, 81)) / 2$$

$$= 95.73 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (165 + \max(13, 120)) / 2$$

$$= 142.30 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= W_{m1} / A_b$$

$$= 1002119 / 979.1981$$

$$= 100.36 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (288460 / 979.1981)$$

$$= 28.89 \text{ N./mm}^2$$



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Flg Calc [Int P]: New Flange

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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	110.73	220.51	164.61	222.00
Radial Flange	8.69	147.00	12.92	148.00
Tangential Flange	80.72	147.00	120.00	148.00
Maximum Average	95.73	147.00	142.30	148.00
Bolting	100.36	151.00	28.89	152.00

Minimum Required Flange Thickness 235.255 mm

Estimated M.A.W.P. (Operating) 6862 KPa.

Estimated Finished Weight of Flange at given Thk. 2118.7 kg.

Estimated Unfinished Weight of Forging at given Thk 2545.3 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	497.36	509.72	0.976	497.36	619.99	0.802
Tangential Flange	St	364.43	382.29	0.953	364.43	464.99	0.784
Tangential Flange	Sto	139.52	254.86	0.547	139.52	309.99	0.450
Combined	Sh/f + St	764.58	764.58	1.000	764.58	929.98	0.822
Combined	Sr + St	403.66	764.58	0.528	403.66	929.98	0.434
Combined	Sh + Sto	636.88	764.58	0.833	636.88	929.98	0.685
Radial Flange	Sr	39.23	509.72	0.077	39.23	619.99	0.063
Maximum Stress Ratio				1.000			0.822



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Flg Calc [Int P]: New Flange

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Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.5426/1.2411)

= 1.2429

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.4092/1.1738) if $h/h0 < 0.35$, else max (0.25; 0.2807/0.7677)

= 0.3487

In this case, the hub ratio $h/h0$, 0.2999 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	461.6 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Flg Calc [Int P]: New Flange

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Root Area of a Single Bolt	Abs	40.7999 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	310.0 N./mm ²
Flange Material Operating Yield Stress	Syo	223.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.455 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$$

$$= \pi/4 (1359.0^2 - 1317.0^2) + (0.5 \cdot 1317.0 \cdot 5.0)$$

$$= 915.6 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207.0 \cdot 915.649 / (24 \cdot 40.8)$$

$$= 193.6 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(193.566; 507.5)$$

$$= 193.6 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(193.566; 290.0)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290.0; 461.599)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 290.0 \geq 140.0(915.649 / (24 \cdot 40.8))$$

$$:: 290.0 \geq 130.9 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290.0 \geq (97.0 \cdot 915.649 + \pi/4 \cdot 5912.596 \cdot 1317.0^2) / (0.7 \cdot 40.8 \cdot 24)$$

$$:: 290.0 \geq 247.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 290.0 \leq 276.0(915.649 / (24 \cdot 40.8))$$

$$:: 290.0 \leq 258.1 \text{ N./mm}^2 \text{ Check Fails}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetafmax})$$

$$:: 290.0 \leq 461.599(1.0/0.455)$$

$$:: 290.0 \leq 1014.7 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 290.0 \cdot 0.2 \cdot 40.8 \cdot 76.2$$

$$= 1838.7 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 77 N./mm²

Bolt Allowable Stress for: SA-193 B16 Sb 151 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



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Flg Calc [Int P]: New Flange

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Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 183893.0 / 1.0 \cdot 9806.637 \cdot 0.361 / (4.386 \cdot 204092 \cdot 31.0^2 \cdot 200.053 \cdot 0.3)$$

$$= 0.658 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 123705.0 / 1.0 \cdot 9806.637 \cdot 0.361 / (4.386 \cdot 176185 \cdot 31.0^2 \cdot 200.053 \cdot 0.3)$$

$$= 0.513 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 183893.0 / 1.0 \cdot 9806.637 \cdot 0.361 / (4.601 \cdot 204092 \cdot 31.0^2 \cdot 200.053 \cdot 0.3)$$

$$= 0.627 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 123705.0 / 1.0 \cdot 9806.637 \cdot 0.361 / (4.601 \cdot 176185 \cdot 31.0^2 \cdot 200.053 \cdot 0.3)$$

$$= 0.489 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.802, Temperature Reduction per Fig. UCS 66.1 = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 7 °C

Min Metal Temp. at Required thickness (UCS 66.1) -4 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	107.46	222.00	158.01 222.00
Radial Flange	9.42	148.00	13.85 148.00
Tangential Flange	76.03	148.00	111.80 148.00
Maximum Average	91.75	148.00	134.90 148.00
Bolting	100.36	152.00	28.89 152.00

Minimum Req'd Thickness, ambient, no Corrosion: 228.346 mm

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Internal Pressure Calculations: Step: 5 9:46am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Blind Flang		5917.1	205	3	1745	147	1
LHS Ch Flg		5912.6	248	3	1285	147	2
LHS Ch Shell		5912.6	34	3	1285	147	3
Main Shell-1		4362.6	24	3	1285	148	4
Dish End		4362.6	28	3	1285	148	5

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Blind Flang		5900	6090.55	6952.5	197	193.153	1
LHS Ch Flg		5900	6848.92	7590.45	240	235.255	2
LHS Ch Shell		5900	6848.9	7590.45	34	29.6069	3
Main Shell-1		4350	4709.79	5406.9	24	22.3713	4
Dish End		4350	4586.67	5278.75	23	21.9678	5

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 4587 KPa.
 Shell Side MAPnc = 5279 KPa.
 Channel Side MAWP = 6091 KPa.
 Channel Side MAPnc = 6869 KPa.

Elements Suitable for Design Internal Pressure.

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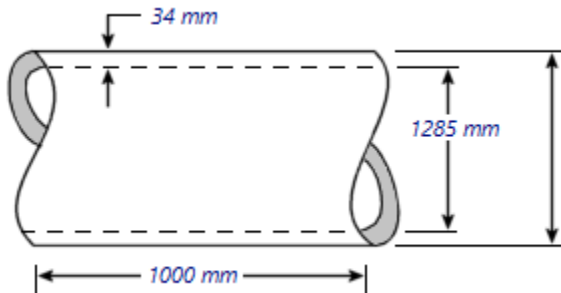
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Element: LHS Ch Shell

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	5912.625 KPa.	Int. Design Temperature	430 °C
Material	SA-387 11	Ext. Design Temperature	200 °C
Length	1000.000 mm	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	147.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 N./mm ²
Inside Diameter Di	1285.000 mm	Actual Stress	126.7 N./mm ²
Spec. Min. Thk t,e, or sa	34.0000 mm	Spec. Nominal Thk.	34.000 mm
Surface Area	42505.8 cm ²	SG of Contents	1.000
Element Volume	1297101.5 cm ³	Weight of Contents	1296.3 Kg
Empty Weight	1091.9 Kg	Operating Weight	2388.2 Kg
		Comp. All. Ope. Stress	82.5 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (5912.625 \cdot 645.5) / (147.0 \cdot 1.0 - 0.6 \cdot 5912.625) \\
 &= 26.6069 + 3.0000 = 29.6069 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 12.625 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (147.0 \cdot 1.0 \cdot 31.0) / (645.5 + 0.6 \cdot 31.0) \\
 &= 6862 - 13 = 6849 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (148.0 \cdot 1.0 \cdot 34.0) / (642.5 + 0.6 \cdot 34.0) \\
 &= 7590 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (5912.625 \cdot (645.5 + 0.6 \cdot 31.0)) / (1.0 \cdot 31.0) \\
 &= 126.671 \text{ N./mm}^2
 \end{aligned}$$



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Element: LHS Ch Shell

Step: 6

9:46am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.578 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 34.0$, $t_r = 27.238$, $c = 3.0$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.879$, Temp. Reduction = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 7 °C

Min Metal Temp. at Required thickness (UCS 66.1) 1 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6090.546 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	34.000 mm	Basic MDMT	7 °C
Stress/Thk. Ratio	0.879	Temperature Difference	7 °C
UCS-66(c) red.	No	Computed Min. Temp.	1 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: LHS Ch Shell

Cylindrical Shell From 30 to Tsht Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
31.000	1353.00	999.97	43.65	0.7391	0.0066822	110.45

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 110.4485) / (3 \cdot 43.6452) = 3373.9385 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.862	1353.00	999.97	350.37	0.7391	0.0002808	27.18

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 27.1793) / (3 \cdot 350.3736) = 103.4239 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
31.000	1353.00	27264.91	43.65	20.1514	0.0005907	57.18

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 57.1823) / (3 \cdot 43.6452) = 1746.7826 \text{ KPa.}$$



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Element: Main Shell-1

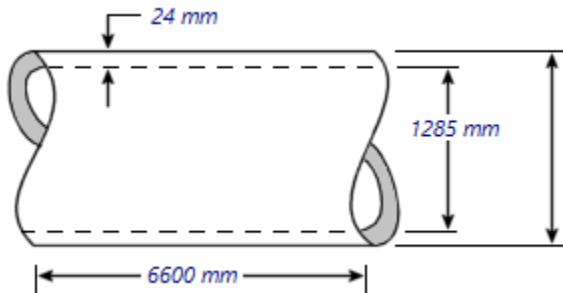
Step: 7

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Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell-1

Cylindrical Shell Design Information:			
Design Pressure P	4362.625 KPa.	Int. Design Temperature	275 °C
Material	SA-387 11	Ext. Design Temperature	275 °C
Length	6600.000 mm	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	148.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 N./mm ²
Inside Diameter Di	1285.000 mm	Actual Stress	136.7 N./mm ²
Spec. Min. Thk t,e, or sa	24.0000 mm	Spec. Nominal Thk.	24.000 mm
Surface Area	276391.1 cm ²	SG of Contents	1.000
Element Volume	7349636.5 cm ³	Weight of Contents	7345.2 Kg
Empty Weight	5048.4 Kg	Operating Weight	12393.5 Kg
		Comp. All. Ope. Stress	104.2 N./mm ²
Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (4362.625 \cdot 645.5) / (148.0 \cdot 1.0 - 0.6 \cdot 4362.625) \\
 &= 19.3713 + 3.0000 = 22.3713 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 12.625 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (148.0 \cdot 1.0 \cdot 21.0) / (645.5 + 0.6 \cdot 21.0) \\
 &= 4722 - 13 = 4710 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (148.0 \cdot 1.0 \cdot 24.0) / (642.5 + 0.6 \cdot 24.0) \\
 &= 5407 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (4362.625 \cdot (645.5 + 0.6 \cdot 21.0)) / (1.0 \cdot 21.0) \\
 &= 136.724 \text{ N./mm}^2
 \end{aligned}$$



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Element: Main Shell-1

Step: 7

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.833 %**Minimum Design Metal Temperature Results:**Govrn. thk, $t_g = 24.0$, $t_r = 20.385$, $c = 3.0$ mm, $E^* = 1.0$ Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.971$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -4 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4586.671 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	0.971	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-4 °C
		Joint Efficiency E^*	100 %

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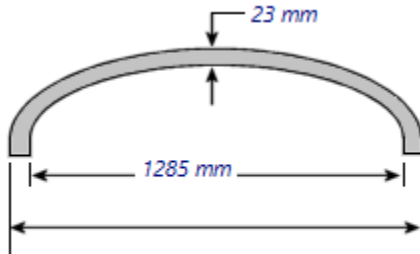
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Element: Dish End

Step: 8 9:46am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Dish End

Elliptical Head Design Information:					
Design Pressure	P	4362.625 KPa.	Int. Design Temperature	275 °C	
Material		SA-387 11	Ext. Design Temperature	275 °C	
Length		50.800 mm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	148.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	148.0 N./mm ²
Inside Diameter	Di	1285.000 mm	Actual Stress	140.4 N./mm ²	
Spec. Min. Thk t,e, or sa		23.0000 mm	Spec. Nominal Thk.	28.000 mm	
Surface Area		21633.2 cm ²	SG of Contents	1.000	
Element Volume		343688.7 cm ³	Weight of Contents	343.5 Kg	
Empty Weight		493.2 Kg	Operating Weight	836.7 Kg	
			Comp. All. Ope. Stress	104.2 N./mm ²	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 16.062 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 8.031 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (4362.625 \cdot 1291.0 \cdot 0.994) / (2 \cdot 148.0 \cdot 1.0 - 0.2 \cdot 4362.625)$$

$$= 18.9678 + 3.0000 = 21.9678 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 12.625 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 148.0 \cdot 1.0 \cdot 20.0) / (0.994 \cdot 1291.0 + 0.2 \cdot 20.0)$$

$$= 4599 - 13 = 4587 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 148.0 \cdot 1.0 \cdot 23.0) / (1.0 \cdot 1285.0 + 0.2 \cdot 23.0)$$

$$= 5279 \text{ KPa.}$$



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Element: Dish End

Step: 8

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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (4362.625 \cdot (0.994 \cdot 1291.0 + 0.2 \cdot 20.0)) / (2 \cdot 1.0 \cdot 20.0)$$

$$= 140.382 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (4362.625 \cdot 645.5) / (148.0 \cdot 1.0 - 0.6 \cdot 4362.625) + 3.0$$

$$= 22.372 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 12.625 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (148.0 \cdot 1.0 \cdot 25.0) / (645.5 + 0.6 \cdot 25.0)$$

$$= 5601 - 13 = 5589 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (1291.0 / (2 \cdot 324.25)))^2 / 6$$

$$= 0.993846$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 9.034 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 23.0, tr = 19.945, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.997$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 16 °C

Min Metal Temp. at Required thickness (UCS 66.1) 16 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 28.0, tr = 20.385, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.815$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 22 °C

Min Metal Temp. at Required thickness (UCS 66.1) 12 °C

[Note:](#)

[Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.](#)

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	4586.671 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	28.000 mm	Basic MDMT	22 °C
Stress/Thk. Ratio	0.815	Temperature Difference	10 °C
UCS-66(c) red.	No	Computed Min. Temp.	12 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 16.062 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 8.031 mm



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Shop Pressure Test Results: Step: 9 9:46am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5962.672 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	5655.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6845.276 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5045.338 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	6220.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	6220.500 KPa.
App 27-4	= M.A.W.P.	4586.671 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7971.183 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7721.800 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	8912.892 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	6744.848 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	8437.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8437.000 KPa.
App 27-4	= M.A.W.P.	6090.546 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	191.2	279.0	0.685	8925.49
Main Shell-1	214.9	279.0	0.770	6857.87
Dish End	220.7	279.0	0.791	6857.87

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	148.00	147.00	1.007
N2-Tube Outlet	Nozzle	148.00	147.00	1.007
Vent	Nozzle	148.00	147.00	1.007
N3-Shell Inlet	Nozzle	148.00	148.00	1.000
N4-Shell Outlet	Nozzle	148.00	148.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind Flang	148.00	147.00	1.007
LHS Ch Flg	148.00	147.00	1.007
LHS Ch Shell	148.00	147.00	1.007
Main Shell-1	148.00	148.00	1.000
Dish End	148.00	148.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
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Shop Pressure Test Results:

Step: 9

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Tube Material	118.00	98.98	1.192
Tubesheet Material	148.00	147.00	1.007

Minimum

1.007

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	59.21	278.99	0.212
N2-Tube Outlet	45.26	278.99	0.162
Vent	43.35	278.99	0.155
N3-Shell Inlet	46.45	278.99	0.166
N4-Shell Outlet	72.95	278.99	0.261

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 10 9:46am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5962.672 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	5655.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6845.276 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5045.338 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	6220.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	6220.500 KPa.
App 27-4	= M.A.W.P.	4586.671 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7971.183 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7721.800 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	8912.892 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	6744.848 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	8437.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8437.000 KPa.
App 27-4	= M.A.W.P.	6090.546 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 4350.0 • 1.0

= 5655.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 5900.0 • 1.007

= 7721.800 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	165.7	279.0	0.594	7734.40
Main Shell-1	177.6	279.0	0.637	5667.60
Dish End	182.4	279.0	0.654	5667.60

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	148.00	147.00	1.007
N2-Tube Outlet	Nozzle	148.00	147.00	1.007
Vent	Nozzle	148.00	147.00	1.007
N3-Shell Inlet	Nozzle	148.00	148.00	1.000
N4-Shell Outlet	Nozzle	148.00	148.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind Flang	148.00	147.00	1.007



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Field Pressure Test Results: Step: 10 9:46am Sep 9, 2026

LHS Ch Flg	148.00	147.00	1.007
LHS Ch Shell	148.00	147.00	1.007
Main Shell-1	148.00	148.00	1.000
Dish End	148.00	148.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	98.98	1.192
Tubesheet Material	148.00	147.00	1.007

Minimum 1.007

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	51.31	278.99	0.184
N2-Tube Outlet	39.22	278.99	0.141
Vent	37.57	278.99	0.135
N3-Shell Inlet	38.39	278.99	0.138
N4-Shell Outlet	60.28	278.99	0.216

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 9:46a Sep 9, 2026

Input Echo, Tubesheet Number 1, Description: U-Tube

Shell Data:

Main Shell Description: Main Shell-1

Shell Test Pressure		5655.00 KPa.
Shell Thickness	ts	24.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1285.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	148.00 N./mm ²
Shell Allowable Stress at Ambient		148.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		7721.80 KPa.
Channel Thickness	tc	34.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1285.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	148.00 N./mm ²
Channel Allowable Stress at Ambient		148.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	626
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	6000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5815.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 T11
Tube Material UNS Number		K11597
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	205.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	29.0000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	618.975 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	4763.3 cm ²
Perimeter of Tube Layout	Cp	3248.0 mm



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ASME TS Calc: Pressure Test

Case: 1 9:46a Sep 9, 2026

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

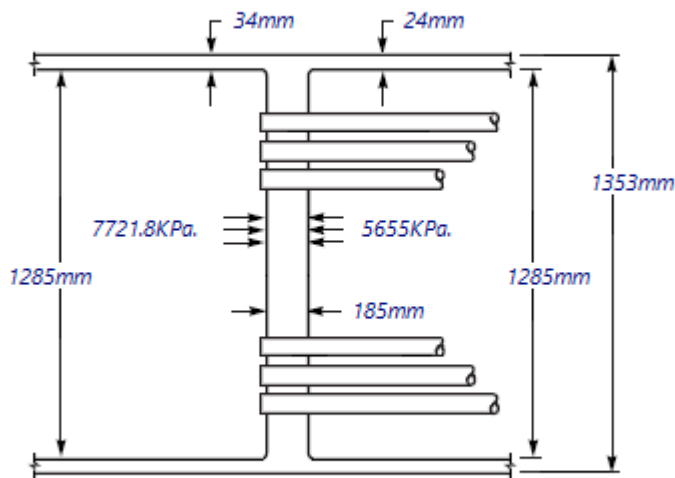
Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	148.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	148.00 N./mm ²
Thickness of Tubesheet	h	185.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1353.000 mm

Area of the Untubed Lanes AL 457.2 cm²

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes
Channel Side Min. Design Pressure	Ptd,min 103.0000 KPa.

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.



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Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TM-1 Material Group C

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(7721.8 \cdot (19.05/2 - 0)) / (118.0 \cdot 1.0 + 0.4 \cdot 7721.8)$

= $0.6075 + 0.0000 = 0.6075$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	CS-2	at	21.11 °C
Elastic Modulus for Material			199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	6433.98	9.03	50.0000	0.0134948	122.73

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 19233.2188$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.7988	19.05	6433.98	23.85	50.0000	0.0019343	103.00

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 102.9979) / (3 \cdot 23.847) = 5758.4932$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.7988 mm
Allowable Internal Pressure at Corroded thickness	28678.85 KPa.
Required Internal Design Pressure	7721.80 KPa.
Allowable External Pressure at Corroded thickness	19233.22 KPa.
Required External Design Pressure	5758.00 KPa.
Required Thickness due to Shell Side pressure	0.7988 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

= $2 \cdot ro + dt$
 = $2 \cdot 618.975 + 19.05$
 = 1257.000 mm

Basic ligament efficiency for shear [μ]:

= $(p - dt) / p$
 = $(29.0 - 19.05) / 29.0$
 = 0.343



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ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 1285.0/1257.0$$

$$= 1.0223$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1285.0/1257.0$$

$$= 1.0223$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/185.0$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t_t(E_t T/E) \cdot (StT/S) \cdot (p); dt - 2t_t)$$

$$= \max(19.05 - 2 \cdot 2.11 (0.20408E+09/.20408E+09) \cdot$$

$$(117/147) \cdot (0); 19.05 - 2 \cdot 2.11)$$

$$= 19.0500 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 29.0 / \sqrt{ (1 - 4 \cdot \min(457.2; 4 \cdot 1257.0 \cdot 29.0) / (\pi \cdot 1257.0^2)) }$$

$$= 29.5494 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (29.5494 - 19.05) / 29.5494$$

$$= 0.3553$$

$h/p = 6.379310$; $\mu^* = 0.355318$; $E^* = 72290128 \text{ KPa}$.

$E^*/E = 0.354224$; $v^* = 0.324368$

Note: As h/p (6.379) is > 2, data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12(1 - v_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (1285.0 + 24.0) 24.0 }$$

$$= 0.0103 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2))$$

$$= 0.01 \cdot 0.20408E+09 \cdot 24.0^3 / (6(1 - 0.3^2))$$

$$= 5299343360.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1285.0 + 24.0) \cdot 0.52993E+1 / 185.0^3 \cdot (1 + 185.0 \cdot 0.01 + (185.0 \cdot 0.01)^2 / 2)$$

$$= 30878272.0 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 1285.0^2 / (4 \cdot 0.20408E+09 \cdot 24.0) \cdot (1 - 1285.0 \cdot 0.3 / (2(1285.0 + 24.0)))$$

$$= 0.71867E-01 \text{ mm/N./mm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$



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$$= 1.0223 \cdot 0.52993E+1 \cdot 0.0103 \cdot 0.071867 (1 + 185.0 \cdot 0.0103)$$

$$= 11569.9150 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1285.0 + 34.0)34.0}$$

$$= 0.0086 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.009 \cdot 0.20408E+09 \cdot 34.0^3 / (6(1 - 0.3^2))$$

$$= 12610686976.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1285.0 + 34.0 / \cos(0)) \cdot 0.12611E+11 / 185.0^3 \cdot (1.0 + 185.0 \cdot 0.009 \cdot (1.0 + 1.0) / 2 + (185.0 \cdot 0.009)^2 / 2 \cdot 1.0)$$

$$= 60669308.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1285.0^2 / (4 \cdot 0.20408E+09 \cdot 34.0) \cdot (1 - 1285.0 \cdot 0.3 / (2(1285.0 + 34.0)))$$

$$= 0.50796E-01 \text{ mm/N} \cdot \text{mm}^2$$

Channel coefficient [zeta α_c]:

$$zeta\alpha_c = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.022 \cdot 0.12611E+11 [0.050796 (1.0 + 0.009 \cdot 1.0 \cdot 185.0) - 0 (1.0 / 185.0 + 1.0 \cdot 0.009 / 2)]$$

$$= 14548.9111 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1353.0 / 1257.0$$

$$= 1.0764$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.32) / 72290128 \cdot (30878272 + 60669308 + 0.20408E+09 \cdot \ln(1.08))$$

$$= 0.9960$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (pc - 1) \cdot (pc^2 + 1) \cdot Pt]$$

$$= 1257.0^2 / 16 \cdot [(1.022 - 1) \cdot (1.022^2 + 1) \cdot 5655.0 - (1.022 - 1) \cdot (1.022^2 + 1) \cdot 7721.8]$$

$$= -9297718.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps$$

$$= -9297718.0 + 14548.911 \cdot 7721.8 - 11569.915 \cdot 5655.0$$

$$= 37618196.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:



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Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M \cdot) - D_o^2/32 \cdot F (P_s - P_t)) / (1 + F)$$

$$= ((37618196) - 1257.0^2/32 \cdot 0.996 (5655.0 - 7721.8)) / (1 + 1.)$$

$$= 69770072.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t)$$

$$= 69770072 + 1257.0^2/64 \cdot (3 + 0.324) \cdot (5655.0 - 7721.8)$$

$$= -99858152.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| M_p | ; | M_o |)$$

$$= \max(| 69770072 | ; | -0.1E+09 |)$$

$$= 99858152.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 99858152 / (0.3553 (185.0 - 8.0)^2)$$

$$= 53.8265 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot S_y$$

$$= 2 \cdot 0.95 \cdot 310.0$$

$$= 588.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 94864568 / (0.3553 (60.2133 - 8.0)^2)$$

$$= 587.6262 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 60.2133 + 0 + 0$$

$$= 60.2133 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(60.2133 ; 3.7499)$$

$$= 60.2133 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| P_s - P_t |$]:

$$= | P_s - P_t |$$

$$= | 5655.0 - 7721.8 |$$

$$= 2066.800 \text{ KPa.}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y$$

$$= 4 \cdot 0.343 \cdot 185.0 / 1257.0 \cdot 0.8 \cdot 0.95 \cdot 309.994$$

$$= 47.587 \text{ N./mm}^2$$

$| P_s - P_t |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.343 \cdot 235.596) \cdot (4 \cdot 4763.3 / 3248.0) \cdot | 5655.0 - 7721.8 | + 0$$

$$= 3.7499 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:



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Note: For a given Shell thickness of: 24.000mm
Min. Shell len. adjacent to the tubesheet is: 316.104 mm

Note: For a given Channel thickness of: 34.000 mm
Min. Channel len. adjacent to the tubesheet is: 376.239mm

Tube Metal Area [At]:
 $= \pi (d_o - t) t$
 $= \pi (19.05 - 2.11) 2.11$
 $= 1.123 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
 $= \pi \cdot t (d_o - t) S$
 $= \pi \cdot 2.11 (19.05 - 2.11) 194.746$
 $= 2229.907 \text{ Kg}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > Ft)
 $= 0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 194.75$
 $= 2229.9067 \text{ Kg}$

Weld Strength Factor [fw]:
 $= S / S_w$ where $S_w = \min(S \text{ and } S_t)$
 $= 194.746 / 194.746$
 $= 1.000$

Design Strength Ratio [fd]:
 $= 1$ for full strength welds

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > Ft)
 $= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 194.75$
 $= 1481.5632 \text{ Kg}$

Minimum Length of Weld [ar]:
 $= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$
 $= 2 \left(\sqrt{(0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) 1.0 \cdot 1.0} - 0.75 \cdot 19.05 \right)$
 $= 2.562 \text{ mm}$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm
 Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 2229.91 is >= tube strength 2229.91 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	53.8 <=	589.0 N./mm ²	Ok
Tubesheet shear stress	4.8 <=	235.6 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	60.213	185.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok



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U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail		
D1uc	106.830	...	218	589	18	236	Ps+Pt D1	Ok
D2uc	91.414	...	167	589	13	236	Ps+Pt D2	Ok
D3uc	60.213	...	54	589	5	236	Ps+Pt D3	Ok
Max:	106.8302	... mm	0.370	0.076	(Str. Ratio)			

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.	Shell	Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	442	...	442	Ok
D2uc	...	...	...	442	...	442	Ok
D3uc	...	...	...	442	...	442	Ok
Max	...	...		...		...	

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	106.830	185.000	Ok
Tube thickness :	0.799	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Given Shell Thickness: 24.0000 mm

Given Channel Thickness: 34.0000 mm

Min Shell length of thk, (24.000) adj. to tubesheet: 316.104 mm

Min Channel length of thk, (34.000) adj. to tubesheet: 376.239 mm

Note: This is a full strength Tube to Tubesheet Joint.



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External Pressure Calculations: Step: 11 9:46am Sep 9,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	194	No Calc	No Calc
20	30	No Calc	...	237	No Calc	No Calc
30	Tsht	999.97	1353	31	0.0066822	110.448
Tsht	40	No Calc	...	31	No Calc	No Calc
40	70	6757.91	1333	21	0.00045995	42.2787
70	80	No Calc	1331	20	0.002087	77.6704

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	197	165.989	103.419	No Calc
20	30	240	235.255	103.419	No Calc
30	Tsht	34	6.86159	103.419	3373.94
Tsht	40	34	No Calc	103.419	No Calc
40	70	24	No Calc	...	888.024
70	80	23	No Calc	...	1296.7

Minimum

888

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	999.97	27264.9	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	70	6757.91	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

Step: 12

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	3799.83	...	3799.83	...	...
20	30	2186.55	402518	2157.57	403247	...
30	40	1091.9	1297101	997.817	1309242	...
40	70	5048.38	7349636	4427.45	7429769	...
70	80	493.248	343689	440.4	348215	...
Total		12619	9392946.00	11823	9490474.00	0



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	...	98.5	...	...	Liquid: 10
10	Insl	155.063	100	...	...	Insulation
20	Liqd	402.273	150	...	...	Liquid: 20
20	Insl	78.0296	140	...	...	Insulation
30	Liqd	1296.31	500	...	...	Liquid: 30
30	Insl	249.604	500	...	...	Insulation
30	Nozl	33.2967	700	688.525	...	N1-Tube Inlet
30	Nozl	23.8051	500	-675.825	...	N2-Tube Outlet
30	Nozl	15.1218	300	663.925	...	Vent
30	Wght	252	500	...	...	Pass partion we
40	Sadl	346.764	1320	912.25	...	Left Saddle flo
40	Sadl	346.764	5280	912.25	...	RH saddle fixed
40	Liqd	7345.15	3300	...	...	Liquid: 40
40	Insl	1047.11	3300	...	...	Insulation
40	Nozl	233.858	3350	792.512	...	N3-Shell Inlet
40	Nozl	267.874	3350	-824.288	...	N4-Shell Outlet
40	Wght	450	1320	-1200	...	Fireproofing
40	Wght	450	5280	-1200	...	Fireproofing we
70	Liqd	343.479	157.883	...	...	Liquid: 70
70	Insl	106.682	25.25	...	...	Insulation
30	FTsh	1805.69	1092.5	...	...	U-Tube
30	Tube	3634.55	4000	...	...	Tubes
40	Baf1	74.0848	568	...	...	Baffles
40	Baf1	74.0848	1136	...	...	Baffles
40	Baf1	74.0848	1704	...	...	Baffles
40	Baf1	74.0848	2272	...	...	Baffles
40	Baf1	74.0848	2840	...	...	Baffles
40	Baf1	74.0848	3408	...	...	Baffles
40	Baf1	74.0848	3976	...	...	Baffles
40	Baf1	74.0848	4544	...	...	Baffles
40	Baf1	74.0848	5112	...	...	Baffles
40	Baf1	74.5093	5680	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	693.5
Liquid	9387.2
Insulation	1636.5



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Element and Detail Weights: Step: 12 9:46am Sep 9,2026

Nozzles	574.0
Weights	1152.0
Exchanger Components	6181.5
Liquid in Tubes	753.6

Sum of the Detail Weights 20378.3 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	12619.9	12619.9	12619.9	12619.9	12619.9	12619.9
Saddles	693.5	693.5	693.5	693.5	693.5	693.5
Nozzles	574.0	574.0	574.0	574.0	574.0	574.0
Wld Weights	252.0	252.0	252.0	252.0	252.0	252.0
Exchanger	6181.5	6181.5	6181.5	6181.5	6181.5	6181.5
Insulation	...	...	...	...	1636.5	1636.5 *
Empty Weights	...	...	...	900.0	900.0	...
Test Weights	...	900.0	...	...	...	...
Ope Weights	...	...	...	...	...	900.0
Ope. Liquid	...	...	...	...	...	9387.2
Tube Ope Lqd	...	...	...	...	...	753.6
Test Liquid	...	9387.2	...	...	...	...
Tube Tst Lqd	...	753.6	...	...	...	...
Totals	20320.9	31361.7	20320.9	21220.9	22857.4	32998.2

Field Installation Options:

* Insulation installed after lifting.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	20320.9 kg.
Shop Test Wt.	- Fabricated weight + water (full)	31361.7 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	20320.9 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	21220.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	22857.4 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	32998.2 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	32201.3 kg.
Field Test Wt.	- Empty weight + test medium (full)	32298.8 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	754048.6 cm ³
Weight of Ope Liq in tubes :	753.6 kg.
Weight of Water in tubes :	753.6 kg.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	35153.8	...
20	30	27156.9	12433.8
30	40	42505.8	40369.5
40	70	276391	266438
70	80	21633.2	19949.8

Total	402840.8	339191.6 cm ²
Total	433.6	365.1 ft ²



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Nozzle Flange MAWP:

Step: 13

9:46am

Sep 9, 2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle	Flange Rating		Des.			Equiv.	- - - -	Max Pressure	
Description	Ope.	Ambient	Temp	Class	Grade/Group	Press	UG-44	50%	DNV

Shellside Flange Rating**Channelside Flange Rating**

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Wind Load Calculation:

Step: 14

9:46am

Sep 9, 2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	9500 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.22 \cdot 0.941)/(1 + 1.7 \cdot 3.4 \cdot 0.22)))$$

$$= \min(0.85, 0.894)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.580
Structure Height to Diameter ratio	5.820

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Z_g = 274320.0$ mm

Effective Height [z]:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 1220.0 + 9500.0 = 10720.0 \text{ mm}$$

$$= 35.171 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\text{Because } z (35.171 \text{ ft.}) > 15 \text{ ft.}$$

$$= 2.01 (z / Z_g)^{2/\alpha}$$

$$= 2.01 (35.171/274320.0)^{2/9.5}$$

$$= 1.016$$

Type of Hill: No Hill



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Wind Load Calculation:

Step: 14

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Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0.0 • 0.0 • 0.0)²

= 1.0

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 1.016 • 1.0 • 1.0 • 1.0 • 60.399²)

= 16.0 psf [78.12] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16.0 • 0.85 • 0.58 • 5.204

= 41.1 lbs. [18.6] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Blind Flang	10720.0	0.000	0.000	0.000	1.016	1.000	78.120
LHS Ch Flg	10720.0	0.000	0.000	0.000	1.016	1.000	78.120
LHS Ch Shell	10720.0	0.000	0.000	0.000	1.016	1.000	78.120
Main Shell-1	10720.0	0.000	0.000	0.000	1.016	1.000	78.120
Dish End	10720.0	0.000	0.000	0.000	1.016	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Pass partion	0.00	10720.00	78.12	0.00
Fireproofing	0.00	9500.00	78.12	0.00
Fireproofing	0.00	9500.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	10720	2454	4834.38	78.12	20.4926
20	30	10720	1878	5634	78.12	23.8821
30	40	10720	1983.6	19836	78.12	84.0833
40	70	10720	1839.6	121414	78.12	514.663
70	80	10720	1835.78	6274.44	78.12	26.5969

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation: Step: 15 9:46am Sep 9,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_n^x$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(6.86520.75)$
 $= 0.085$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33.0)$
 $= 0.030$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.085, 0.03)$ per 12.8.2
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.157 \cdot 32305 \cdot 1.25$
 $= 1899.514$ Kgf

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.157 \cdot 32305$
 $= 1013.1$ Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	872.5	4614.95	350.571



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Earthquake Load Calculation: Step: 15 9:46am Sep 9,2026

20	30	642.5	4614.95	258.157
30	40	642.5	4614.95	258.157
40	Sad1	642.5	4614.95	258.157
Sad1	70	642.5	4614.95	258.157
40	70	642.5	4614.95	258.157
70	80	642.5	4614.95	258.157

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 16 9:46am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	4988.351 mm
Center of Gravity of the Liquid	4365.948 mm
Center of Gravity of the Insulation	3912.644 mm
Center of Gravity of the Nozzles	4536.939 mm
Center of Gravity of the Added Weights (Operating)	4116.631 mm
Center of Gravity of the Added Weights (Empty)	4116.631 mm
Center of Gravity of the Tubesheet(s)	1595.850 mm
Center of Gravity of the Tubes	4503.350 mm
Center of Gravity of the Baffles	4813.815 mm
Center of Gravity of Bare Shell New and Cold	2506.473 mm
Center of Gravity of Bare Shell Corroded	2366.675 mm
Vessel CG in the Operating Condition	3490.090 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	3075.033 mm
Vessel CG in the Test Condition	3496.101 mm

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Saddle Calcs: Road Transport:

Step: 17

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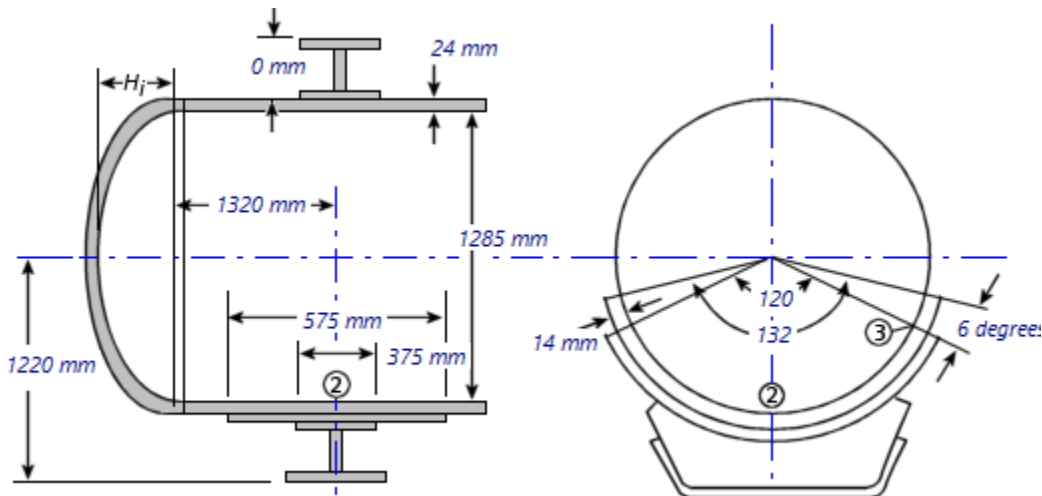
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	654.50	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	6650.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1320.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	575.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	148.00	N./mm ²
Shell Allowable Stress used in Calculation		148.00	N./mm ²
Head Allowable Stress used in Calculation		147.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1220.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		26953.29	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	333.7	Km/hr



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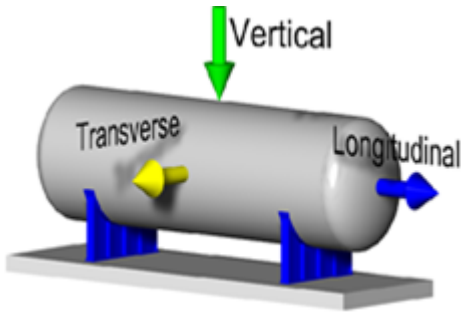
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Saddle Calcs: Road Transport:

Step: 17

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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-3.07	-278.99
Long. Stress at Top of Midspan	3.07	148.00
Long. Stress at Bottom of Midspan	3.07	148.00
Long. Stress at Top of Saddles	17.64	148.00
Long. Stress at Bottom of Saddles	-9.78	-278.99
Long. Stress at Bottom of Saddles	9.78	148.00
Tangential Shear in Shell	11.88	118.40
Circ. Stress at Horn of Saddle	17.56	222.00
Circ. Compressive Stress in Shell	0.93	148.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 170.06 \cdot 0.00256 \cdot 207.37^2 \cdot 1.0 \\
 &= 18721.1 \text{ lbf} \\
 &= 8491.911 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 5.358^2/4 \cdot 0.00256 \cdot 207.37^2 \cdot 1.0 \cdot 1.2 \\
 &= 2978.1 \text{ lbf} \\
 &= 1350.9 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 19627.37 \cdot 1.0 + 8491.91 \\
 &= 10454.6 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 19627.37 \cdot 1.0 + 1350.87 \\
 &= 3313.6 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 10454.6/2 \cdot 1220.0/1178.6606 \\
 &= 5410.7 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 3314 \cdot 1220.0/3960.001 \\
 &= 1020.9 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 20516.8 \cdot 0.05
 \end{aligned}$$



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Saddle Calcs: Road Transport: Step: 17 9:46am Sep 9, 2026

= 1025.840 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= 20517 + 1026 + max(1021, 5411)

= 26953.3 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 27300.06 Kgf

Transverse Shear Load 5227.32 Kgf

Longitudinal Shear Load 3313.60 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(1285.0/2 + 24.0 + 14.0)sin(120.0/2), 2•654.5)

= 1178.661 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -26953•1320.0[1-(1-1320.0/6650.8+(654.5²-0.0²)/

(2•1320.0•6650.8))/(1+(4•0.0)/(3•6650.8))]

= -6193.4 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 26953•6651/4((1+2(655²-0²)/(6651²))/(1+(4•0)/

(3•6651))-4•1320/6651)

= 10105.1 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0.0 • 654.5/(2•24.0) - 10105.1/(π•654.5²•24.0)

= -3.07 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0.0 • 654.5/(2 • 24.0) + 10105.1/(π • 654.5² • 24.0)

= 3.07 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)

= 0.0•654.5/(2•24.0)--6193.4/(0.1066•π•654.5²•24.0)

= 17.64 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 0.0•654.5/(2•24.0)+6193.4/(0.1923•π•654.5²•24.0)

= -9.78 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 26953(6650.8 - 2 • 1320.0)/(6650.8 + (4 • 0.0/3))

= 16254.3 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 16254.33 / (654.5 \cdot 24.0)$$

$$= 11.88 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(654.5 \cdot 24.0)}$$

$$= 97.759 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(375.0 + 1.56 \cdot \sqrt{(654.5 \cdot 24.0)}, 2 \cdot 1320.0)$$

$$= 570.52 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 26953 \cdot 0.1 / (570.517(24.0 + 1.0 \cdot 14.0))$$

$$= -0.93 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -26953 / (4(24.0 + 1.0 \cdot 14.0)570.517) -$$

$$3 \cdot 0.053 \cdot 26953 / (2(24.0 + 1.0 \cdot 14.0)^2)$$

$$= -17.56 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1400.0000	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	N./mm ²
Height of Web at Center	Hw, c	560.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(642.5 \cdot 24.0)} + 575.0)24.0^3) / 12$$

$$= 88.556 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 88.556 + 184.492(172.748 - 12.0)^2$$

$$= 47760.867 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	768.7	24.0	12.0	184.5	221390.4	0.478E+05
Wearplate	575.0	14.0	31.0	80.5	249550.0	0.162E+05
Web	10.0	517.5	296.8	51.8	1535681.1	0.195E+05
BasePlate	400.0	22.0	566.5	88.0	4985199.5	0.136E+06
Totals	...	...	...	404.7	6991821.0	0.220E+06

Distance to Centroid [C1]:

$$= AY / A$$



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Saddle Calcs: Road Transport: Step: 17 9:46am Sep 9,2026

$$= 6991821/404.742$$

$$= 172.748 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 26953.295$$

$$= 5485.5796 \text{ Kg}$$

Tension Stress, St = (Fh/As) = 2.4425 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 185.9966 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1220.0 - 642.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 688.657 \text{ mm}$$

Bending Moment, M = Fh * d = 3777.7561 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 2.9100 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 206.6629 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(26953 + 347)400.0 / (4 \cdot 1400.0 \cdot 206.663)}$$

$$= 16.661 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (642.5 + 24.0 + 14.0)$$

$$= 1178.661 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1178.6606 / (4 - 1)$$

$$= 392.887 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 392.8869 \cdot 400.0/2$$

$$= 785.774 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 27300.057 / (1400.0 \cdot 400.0)$$

$$= 4.875 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 785.8 \cdot 4.88$$

$$= 3830.655 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



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$$= 36.5 + 19.644$$

$$= 56.144 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 3830.7/56.1443$$

$$= 6.691 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 375.0 - 10.0$$

$$= 365.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/56.144$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375.0/2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4394.5/56.144)}$$

$$= 88.471 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1220.0 - \cos(120.0/2) (642.5 + 24.0 + 14.0) - 22.0$$

$$= 857.750 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/310.0)}$$

$$= 112.838$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 857.75/88.471$$

$$= 9.695$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 3313.6 / (2 \cdot 1400.0) \cdot 392.887 \cdot 857.75/2$$

$$= 199.411 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($9.6952 < 112.8376$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.7)^2 / (2 \cdot 112.84^2)) 310 /$$

$$(5/3 + 3 \cdot (9.7) / (8 \cdot 112.84) - (9.7^3) / (8 \cdot 112.84^3))$$

$$= 181.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 6.69/181.8 + (199.41 \cdot 187.5/43945308) / 206.66$$



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= 0.077

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	392.9	10.0	0.0	39.3	0.0	3.27
Totals	...	...	...	75.8	...	0.440E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/75.789

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 375.0/2

= 187.500 mm

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk

= 1220.0 - $\sqrt{(680.5 + 14.0)^2 - (392.887/2)^2}$ - 22.0

= 546.471 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$

= $\sqrt{(4397.7/75.789)}$

= 76.175 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 546.471/76.175

= 7.174

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 3313.604/(2 • 1400.0)

= 1.183 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 1.183 • 392.887 • 546.471

= 254.089 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 254.089/234545.141

= 10.624 N./mm²

Compressive Allowable, KL/r < Cc (7.1739 < 112.8376) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (7.17)^2 / (2 \cdot 112.84^2)) 310 /$

$(5/3 + 3 \cdot (7.17) / (8 \cdot 112.84) - (7.17^3) / (8 \cdot 112.84^3))$

= 183 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 9.92/183.01 + (254.09 • 187.5/4397.717)/206.66

= 0.106

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



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Vessel Mean Radius	Rm	654.50	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	6650.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1320.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	575.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	148.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	332.75	mm
Shell Allowable Stress used in Calculation		148.00	N./mm ²
Head Allowable Stress used in Calculation		148.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1220.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		5421.77	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	333.7	Km/hr

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-0.43	-278.99
Long. Stress at Top of Midspan	0.43	148.00
Long. Stress at Bottom of Midspan	0.43	148.00
Long. Stress at Top of Saddles	4.72	148.00
Long. Stress at Bottom of Saddles	-2.62	-278.99
Long. Stress at Bottom of Saddles	2.62	148.00
Tangential Shear in Shell	2.24	118.40
Circ. Stress at Horn of Saddle	3.53	222.00
Circ. Compressive Stress in Shell	0.19	148.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 170.06 \cdot 0.00256 \cdot 207.37^2 \cdot 1.0 \\
 &= 18721.1 \text{ lbf} \\
 &= 8491.911 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 5.358^2/4 \cdot 0.00256 \cdot 207.37^2 \cdot 1.0 \cdot 1.2 \\
 &= 2978.1 \text{ lbf} \\
 &= 1350.9 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:



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$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 19627.37 \cdot 1.0 + 8491.91 \\
 &= 10454.6 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [FI]:

$$\begin{aligned}
 &= gx \cdot \text{Shipping Weight} \cdot gw + Flw \\
 &= 0.1 \cdot 19627.37 \cdot 1.0 + 1350.87 \\
 &= 3313.6 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [Fwt]:

$$\begin{aligned}
 &= Ft/\text{Num of Saddles} \cdot B / E \\
 &= 10454.6/2 \cdot 1220.0/1178.6606 \\
 &= 5410.7 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [Fwl]:

$$\begin{aligned}
 &= FI \cdot B / Ls \\
 &= 3314 \cdot 1220.0/3960.001 \\
 &= 1020.9 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot gy \\
 &= 10.6 \cdot 0.05 \\
 &= 0.529 \text{ Kgf}
 \end{aligned}$$

Load Combination Results due to Transport Loading [Q]:

$$\begin{aligned}
 &= \text{Saddle Load} + Frv + \max(Fwl, Fwt) \\
 &= 11 + 0.5 + \max(1021, 5411) \\
 &= 5421.8 \text{ Kgf}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	5768.54	Kgf
Transverse Shear Load	5227.32	Kgf
Longitudinal Shear Load	3313.60	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm) \\
 &= \min(2(1285.0/2 + 24.0 + 14.0) \sin(120.0/2), 2 \cdot 654.5) \\
 &= 1178.661 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm^2)/3L)] \\
 &= -5422 \cdot 1320.0 [1 - (1 - 1320.0/6650.8 + (654.5^2 - 332.75^2)/(2 \cdot 1320.0 \cdot 6650.8))/(1 + (4 \cdot 332.75^2)/(3 \cdot 6650.8))] \\
 &= -1657.8 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L) \\
 &= 5422 \cdot 6651/4((1 + 2(655^2 - 333^2)/(6651^2))/(1 + (4 \cdot 333)/(3 \cdot 6651)) - 4 \cdot 1320/6651) \\
 &= 1415.7 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$



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$$= 0.0 \cdot 654.5 / (2 \cdot 24.0) - 1415.7 / (\pi \cdot 654.5^2 \cdot 24.0)$$

$$= -0.43 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 654.5 / (2 \cdot 24.0) + 1415.7 / (\pi \cdot 654.5^2 \cdot 24.0)$$

$$= 0.43 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 654.5 / (2 \cdot 24.0) - 1657.8 / (0.1066 \cdot \pi \cdot 654.5^2 \cdot 24.0)$$

$$= 4.72 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 654.5 / (2 \cdot 24.0) + 1657.8 / (0.1923 \cdot \pi \cdot 654.5^2 \cdot 24.0)$$

$$= -2.62 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 5422(6650.8 - 2 \cdot 1320.0) / (6650.8 + (4 \cdot 332.75 / 3))$$

$$= 3065.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 3065.16 / (654.5 \cdot 24.0)$$

$$= 2.24 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(654.5 \cdot 24.0)}$$

$$= 97.759 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(375.0 + 1.56 \cdot \sqrt{(654.5 \cdot 24.0)}, 2 \cdot 1320.0)$$

$$= 570.52 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 5422 \cdot 0.1 / (570.517(24.0 + 1.0 \cdot 14.0))$$

$$= -0.19 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -5422 / (4(24.0 + 1.0 \cdot 14.0)570.517) - 3 \cdot 0.053 \cdot 5422 / (2(24.0 + 1.0 \cdot 14.0)^2)$$

$$= -3.53 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1400.0000	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	N./mm ²
Height of Web at Center	Hw,c	560.0	mm
Friction Coefficient	mu	0.000	



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Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(642.5 \cdot 24.0) + 575.0})^3) / 12$$

$$= 88.556 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 88.556 + 184.492 (172.748 - 12.0)^2$$

$$= 47760.867 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	768.7	24.0	12.0	184.5	221390.4	0.478E+05
Wearplate	575.0	14.0	31.0	80.5	249550.0	0.162E+05
Web	10.0	517.5	296.8	51.8	1535681.1	0.195E+05
BasePlate	400.0	22.0	566.5	88.0	4985199.5	0.136E+06
Totals	...	...	...	404.7	6991821.0	0.220E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6991821 / 404.742$$

$$= 172.748 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 5421.772$$

$$= 1103.4482 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.4913 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 185.9966 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1220.0 - 642.5 \cdot \sin(120.0 / 2) / 1.0472$$

$$= 688.657 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 759.9121 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.5854 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 206.6629 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(5422 + 347) \cdot 400.0) / (4 \cdot 1400.0 \cdot 206.663)}$$

$$= 7.659 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0 / 2) (642.5 + 24.0 + 14.0)$$

$$= 1178.661 \text{ mm}$$



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Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1178.6606 / (4 - 1)$$

$$= 392.887 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 392.8869 \cdot 400.0 / 2$$

$$= 785.774 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 5768.536 / (1400.0 \cdot 400.0)$$

$$= 1.030 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 785.8 \cdot 1.03$$

$$= 809.422 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 36.5 + 19.644$$

$$= 56.144 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 809.4 / 56.1443$$

$$= 1.414 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 375.0 - 10.0$$

$$= 365.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0.0 / 56.144$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375.0 / 2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4394.5 / 56.144)}$$

$$= 88.471 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1220.0 - \cos(120.0/2) (642.5 + 24.0 + 14.0) - 22.0$$

$$= 857.750 \text{ mm}$$

Intermediate Term [Cc]:



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$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 310.0)}$$

$$= 112.838$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 857.75 / 88.471$$

$$= 9.695$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 3313.6 / (2 \cdot 1400.0) \cdot 392.887 \cdot 857.75 / 2$$

$$= 199.411 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($9.6952 < 112.8376$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.7)^2 / (2 \cdot 112.84^2)) 310 /$$

$$(5/3 + 3 \cdot (9.7) / (8 \cdot 112.84) - (9.7^3) / (8 \cdot 112.84^3))$$

$$= 181.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 1.41 / 181.8 + (199.41 \cdot 187.5 / 43945308) / 206.66$$

$$= 0.048$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	392.9	10.0	0.0	39.3	0.0	3.27
Totals	...	...	...	75.8	...	0.440E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 75.789$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375.0 / 2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (\text{Pitch}/2)^2} - B_{pthk}$$

$$= 1220.0 - \sqrt{(680.5 + 14.0)^2 - (392.887/2)^2} - 22.0$$

$$= 546.471 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4397.7 / 75.789)}$$

$$= 76.175 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 546.471 / 76.175$$

$$= 7.174$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 3313.604 / (2 \cdot 1400.0)$$

$$= 1.183 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:



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= Unit Force • e • L
 = 1.183 • 392.887 • 546.471
 = 254.089 Kg-m.

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

= Bending Moment / Section Modulus
 = 254.089/234545.141
 = 10.624 N./mm²

Compressive Allowable, $KL/r < C_c$ (7.1739 < 112.8376) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)
 = (1-(7.17)²/(2 • 112.84²))310/
 (5/3+3•(7.17)/(8• 112.84)-(7.17³)/(8•112.84³)
 = 183 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} • C_{1/I})/S_{ba}$
 = 2.1/183.01 + (254.09 • 187.5/4397.717)/206.66
 = 0.063

Review notes about nozzle loadings and supports in the Warnings report.

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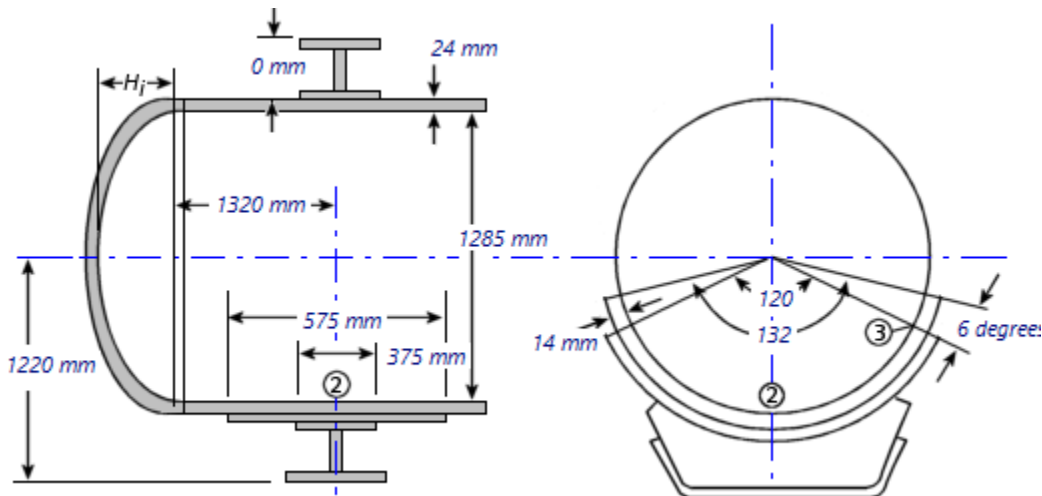
Saddle Calcs: Operating Case: Step: 18 9:46am Sep 9,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	656.00	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	6650.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1320.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	575.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	148.00	N./mm ²
Shell Allowable Stress used in Calculation		148.00	N./mm ²
Head Allowable Stress used in Calculation		147.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1220.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	31873.42	Kgf
Pressure used in Analysis	P	4356.298 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	63.92	148.00
Long. Stress at Bottom of Midspan	72.17	148.00
Long. Stress at Top of Saddles	91.76	148.00
Long. Stress at Bottom of Saddles	54.90	148.00
Tangential Shear in Shell	16.02	118.40
Circ. Stress at Horn of Saddle	24.23	185.00
Circ. Compressive Stress in Shell	1.22	148.00



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(669.7/2 + 0) \cdot 1220.0/1178.6606$$

$$= 1039.8 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(259.17, 0) \cdot 1220.0/3960.001$$

$$= 79.8 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(1899.51, 0) \cdot 1220.0/3960.001$$

$$= 585.2 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(1900/2 + 0) \cdot 1220.0/1178.6606$$

$$= 2949.2 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 28924 + \max(80, 1040, 585, 2949)$$

$$= 31873.4 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(0.0 + \pi/4(1.633 \cdot 1.2)^2)$$

$$= 2541.528 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	32220.18	Kgf
Transverse Shear Load	949.76	Kgf
Longitudinal Shear Load	1899.51	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(1285.0/2 + 24.0 + 14.0)\sin(120.0/2), 2 \cdot 656.0)$$

$$= 1178.661 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -31873 \cdot 1320.0 [1 - (1 - 1320.0/6650.8 + (656.0^2 - 0.0^2)/(2 \cdot 1320.0 \cdot 6650.8))/(1 + (4 \cdot 0.0)/(3 \cdot 6650.8))]$$

$$= -7319.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 31873 \cdot 6651/4((1 + 2(656^2 - 0^2)/(6651^2))/(1 + (4 \cdot 0)/$$



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$$(3 \cdot 6651) - 4 \cdot 1320 / 6651 \\ = 11954.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t) \\ = 4356.298 \cdot 656.0 / (2 \cdot 21.0) - 11954.4 / (\pi \cdot 656.0^2 \cdot 21.0) \\ = 63.92 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 4356.298 \cdot 656.0 / (2 \cdot 21.0) + 11954.4 / (\pi \cdot 656.0^2 \cdot 21.0) \\ = 72.17 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t) \\ = 4356.298 \cdot 656.0 / (2 \cdot 21.0) - 7319.3 / (0.1066 \cdot \pi \cdot 656.0^2 \cdot 21.0) \\ = 91.76 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 4356.298 \cdot 656.0 / (2 \cdot 21.0) + 7319.3 / (0.1923 \cdot \pi \cdot 656.0^2 \cdot 21.0) \\ = 54.90 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 31873(6650.8 - 2 \cdot 1320.0) / (6650.8 + (4 \cdot 0.0 / 3)) \\ = 19221.4 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t) \\ = 1.1707 \cdot 19221.43 / (656.0 \cdot 21.0) \\ = 16.02 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{656.0 \cdot 21.0} \\ = 91.550 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(375.0 + 1.56 \cdot \sqrt{656.0 \cdot 21.0}, 2 \cdot 1320.0) \\ = 558.10 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1) \\ = \min(148.0 / 148.0, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr)) \\ = -0.7603 \cdot 31873 \cdot 0.1 / (558.099(21.0 + 1.0 \cdot 14.0)) \\ = -1.22 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ = -31873 / (4(21.0 + 1.0 \cdot 14.0)558.099) - 3 \cdot 0.053 \cdot 31873 / (2(21.0 + 1.0 \cdot 14.0)^2) \\ = -24.23 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 3960.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.000013 \cdot 3960.001(275.0 - 21.1)$$



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= 13.210 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1400.0000	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	249.0	N./mm ²
Height of Web at Center	Hw,c	560.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(645.5 \cdot 21.0)} + 575.0) 21.0^3) / 12$$

$$= 58.393 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 58.393 + 158.892 (181.23 - 10.5)^2$$

$$= 46373.379 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	756.6	21.0	10.5	158.9	166836.4	0.464E+05
Wearplate	575.0	14.0	28.0	80.5	225400.0	0.189E+05
Web	10.0	517.5	293.8	51.8	1520156.1	0.181E+05
BasePlate	400.0	22.0	563.5	88.0	4958799.5	0.129E+06
Totals	...	...	...	379.1	6871192.0	0.212E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6871192 / 379.142$$

$$= 181.230 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 31873.418$$

$$= 6486.9316 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.8884 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 149.4000 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1220.0 - 645.5 \cdot \sin(120.0 / 2) / 1.0472$$

$$= 686.176 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 4451.2632 \text{ Kg-m.}$$



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Bending Stress, $S_b = (M * C_1 / I) = 3.7313 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 * \text{Yield Str} = 166.0000 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(31873 + 347)400.0 / (4 * 1400.0 * 166.0))}$
 $= 20.196 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120.0/2) (642.5 + 24.0 + 14.0)$
 $= 1178.661 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1178.6606 / (4 - 1)$
 $= 392.887 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 392.8869 * 400.0 / 2$
 $= 785.774 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 32220.182 / (1400.0 * 400.0)$
 $= 5.754 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p * B_p$
 $= 785.8 * 5.75$
 $= 4521.031 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 36.5 + 19.644$
 $= 56.144 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 4521.0 / 56.1443$
 $= 7.897 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 375.0 - 10.0$
 $= 365.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0 / 56.144$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



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= Saddle Width / 2
= 375.0/2
= 187.500 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{4394.5/56.144}$
= 88.471 mm

Length of Outer Rib [L]:
= Saddle Height - $\cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
= 1220.0 - $\cos(120.0/2) (642.5 + 24.0 + 14.0) - 22.0$
= 857.750 mm

Intermediate Term [Cc]:
= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/249.0}$
= 125.902

Slenderness ratio [KL/r]:
= KL/r
= 1 • 857.75/88.471
= 9.695

Bending Moment [Rm]:
= FI / (2 • Bplen) • e • L / 2
= 1899.5 / (2 • 1400.0) • 392.887 • 857.75/2
= 114.312 Kg-m.

Compressive Allowable, $KL/r < Cc$ (9.6952 < 125.9015) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (9.7)^2 / (2 \cdot 125.9^2)) 249 / (5/3 + 3 \cdot (9.7) / (8 \cdot 125.9) - (9.7)^3 / (8 \cdot 125.9^3))$
= 146.4 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
= Sc/Sca + (Rm • C1 / I)/Sba
= 7.9/146.43 + (114.31 • 187.5/43945308)/166.0
= 0.083

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	392.9	10.0	0.0	39.3	0.0	3.27
Totals	...	...	...	75.8	...	0.440E+04

Distance to Centroid from Datum [ytot]:
= AY / A
= 0.0/75.789
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 375.0/2
= 187.500 mm

Length of Inner Rib [L]:
= Saddle Height - $\sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$
= 1220.0 - $\sqrt{(680.5 + 14.0)^2 - (392.887/2)^2} - 22.0$
= 546.471 mm

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{(4397.7 / 75.789)}$$

$$= 76.175 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 546.471 / 76.175$$

$$= 7.174$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1899.514 / (2 \cdot 1400.0)$$

$$= 0.678 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.678 \cdot 392.887 \cdot 546.471$$

$$= 145.656 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 145.656 / 234545.141$$

$$= 6.090 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.1739 < 125.9015$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.17)^2 / (2 \cdot 125.9^2)) 249 /$$

$$(5/3 + 3 \cdot (7.17) / (8 \cdot 125.9) - (7.17)^3 / (8 \cdot 125.9^3))$$

$$= 147.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 11.71 / 147.27 + (145.66 \cdot 187.5 / 4397.717) / 166.0$$

$$= 0.116$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	148.00 N./mm ²
Area Available in a Single Bolt	BlArea	3.0428 cm ²
Saddle Load Q0 (Weight)	Q0	29271.0 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	585.2 Kg
Maximum Transverse Force	Ft	949.8 Kg
Maximum Longitudinal Force	F1	1899.5 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1899.5 / (3.043 \cdot 6)$$

$$= 10.2 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 949.8 / (3.043 \cdot 6)$$

$$= 5.1 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (BoltShearAllowable \cdot Nbolts)$
= $1899.51 / (82.8 \cdot 6.0)$
= 0.3750 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1220.0 \cdot 949.76 + 0.0$
= 1158.73 Kg-m.

Eccentricity (e):
= $Rmom / QO$
= $1158.73 / 29270.98$
= 39.59 mm < $Bplen / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0.0, 0.375, 0.0]$
= 0.3750 cm²

MDMT Calculations for: Left Saddle flo, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $tg = 14.0$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.971$, Temp. Reduction = 2 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
Min Metal Temp. at Required thickness (UCS 66.1) -20 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	656.00	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	6650.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1320.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	575.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	148.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	334.25	mm
Shell Allowable Stress used in Calculation		148.00	N./mm ²
Head Allowable Stress used in Calculation		148.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1220.00	mm
Coefficient of Friction	mu	0.00	



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Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 7342.72 Kgf
Pressure used in Analysis P 4356.298 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	67.38	148.00
Long. Stress at Bottom of Midspan	68.71	148.00
Long. Stress at Top of Saddles	75.32	148.00
Long. Stress at Bottom of Saddles	64.01	148.00
Tangential Shear in Shell	3.46	118.40
Circ. Stress at Horn of Saddle	5.58	185.00
Circ. Compressive Stress in Shell	0.28	148.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= $F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
= $3.0 (669.7/2 + 0) \cdot 1220.0/1178.6606$
= 1039.8 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
= $\max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
= $\max(259.17, 0) \cdot 1220.0/3960.001$
= 79.8 Kgf

Saddle Reaction Force due to Earthquake Fl [Fsl]:
= $\max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
= $\max(1899.51, 0) \cdot 1220.0/3960.001$
= 585.2 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:
= $F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
= $3.0 (1900/2 + 0) \cdot 1220.0/1178.6606$
= 2949.2 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
= Saddle Load + $\max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
= $4394 + \max(80, 1040, 585, 2949)$
= 7342.7 Kgf

Longitudinal Wind Force [Fl]:
= $\text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
= $1.1 \cdot 766.08 (0.0 + \pi/4 (1.633 \cdot 1.2)^2)$
= 2541.528 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	7689.48	Kgf
Transverse Shear Load	949.76	Kgf
Longitudinal Shear Load	1899.51	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
= $\min(2 (ShellID/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
= $\min(2 (1285.0/2 + 24.0 + 14.0) \sin(120.0/2), 2 \cdot 656.0)$
= 1178.661 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405



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K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -7343 \cdot 1320.0 \left[1 - \left(1 - \frac{1320.0}{6650.8} + \frac{(656.0^2 - 334.25^2)}{(2 \cdot 1320.0 \cdot 6650.8)} \right) / \left(1 + \frac{(4 \cdot 334.25)}{(3 \cdot 6650.8)} \right) \right]$$

$$= -2246.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L}{(3 \cdot 6651)} \right) - 4 \cdot 1320 / 6651$$

$$= 7343 \cdot 6651 / 4 \left(\frac{(1 + 2(656^2 - 334^2)/(6651^2)) / (1 + (4 \cdot 334)/(3 \cdot 6651)) - 4 \cdot 1320 / 6651}{(3 \cdot 6651)} \right) - 4 \cdot 1320 / 6651$$

$$= 1914.5 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 4356.298 \cdot 656.0 / (2 \cdot 21.0) - 1914.5 / (\pi \cdot 656.0^2 \cdot 21.0)$$

$$= 67.38 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 4356.298 \cdot 656.0 / (2 \cdot 21.0) + 1914.5 / (\pi \cdot 656.0^2 \cdot 21.0)$$

$$= 68.71 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 4356.298 \cdot 656.0 / (2 \cdot 21.0) - 2246.8 / (0.1066 \cdot \pi \cdot 656.0^2 \cdot 21.0)$$

$$= 75.32 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 4356.298 \cdot 656.0 / (2 \cdot 21.0) + 2246.8 / (0.1923 \cdot \pi \cdot 656.0^2 \cdot 21.0)$$

$$= 64.01 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 7343(6650.8 - 2 \cdot 1320.0) / (6650.8 + (4 \cdot 334.25 / 3))$$

$$= 4150.0 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 4149.98 / (656.0 \cdot 21.0)$$

$$= 3.46 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(656.0 \cdot 21.0)}$$

$$= 91.550 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(375.0 + 1.56 \cdot \sqrt{(656.0 \cdot 21.0)}, 2 \cdot 1320.0)$$

$$= 558.10 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η_a]:

$$= \min(Sr/S, 1)$$

$$= \min(148.0/148.0, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:



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$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.7603 \cdot 7343 \cdot 0.1 / (558.099(21.0 + 1.0 \cdot 14.0)) \\
 &= -0.28 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) $[\sigma_r, r]$:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2) \\
 &= -7343 / (4(21.0 + 1.0 \cdot 14.0)558.099) - \\
 &\quad 3 \cdot 0.053 \cdot 7343 / (2(21.0 + 1.0 \cdot 14.0)^2) \\
 &= -5.58 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1400.0000	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	249.0	N./mm ²
Height of Web at Center	Hw,c	560.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$\begin{aligned}
 &= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12 \\
 &= ((1.56 \cdot \sqrt{(645.5 \cdot 21.0)} + 575.0)21.0^3) / 12 \\
 &= 58.393 \text{ cm}^4
 \end{aligned}$$

Resolved Inertia for the Shell [I,shell]:

$$\begin{aligned}
 &= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2 \\
 &= 58.393 + 158.892(181.23 - 10.5)^2 \\
 &= 46373.379 \text{ cm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	756.6	21.0	10.5	158.9	166836.4	0.464E+05
Wearplate	575.0	14.0	28.0	80.5	225400.0	0.189E+05
Web	10.0	517.5	293.8	51.8	1520156.1	0.181E+05
BasePlate	400.0	22.0	563.5	88.0	4958799.5	0.129E+06
Totals	...	...	...	379.1	6871192.0	0.212E+06

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 6871192 / 379.142 \\
 &= 181.230 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle} / 2 \\
 &= 180 - 120.0 / 2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta)) \\
 &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.204 \cdot 7342.72 \\
 &= 1494.4027 \text{ Kg}
 \end{aligned}$$



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Tension Stress, $St = (Fh/As) = 0.6654 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 * \text{Yield Str} = 149.4000 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1220.0 - 645.5 \cdot \sin(120.0/2) / 1.0472$
 $= 686.176 \text{ mm}$

Bending Moment, $M = Fh \cdot d = 1025.4432 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 0.8596 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 * \text{Yield Str} = 166.0000 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(7343 + 347) \cdot 400.0) / (4 \cdot 1400.0 \cdot 166.0)}$
 $= 9.866 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120.0/2) \cdot (642.5 + 24.0 + 14.0)$
 $= 1178.661 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1178.6606 / (4 - 1)$
 $= 392.887 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 392.8869 \cdot 400.0 / 2$
 $= 785.774 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 7689.483 / (1400.0 \cdot 400.0)$
 $= 1.373 \text{ Kg/cm}^2$

Axial Load [P]:
 $= Ap \cdot Bp$
 $= 785.8 \cdot 1.37$
 $= 1078.963 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 36.5 + 19.644$
 $= 56.144 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 1079.0 / 56.1443$
 $= 1.885 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$



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= 375.0 - 10.0
= 365.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
= 0.0/56.144
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
= 375.0/2
= 187.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{4394.5/56.144}$
= 88.471 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 1220.0 - cos(120.0/2)(642.5 + 24.0 + 14.0) - 22.0
= 857.750 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/249.0}$
= 125.902

Slenderness ratio [KL/r]:

= KL/r
= 1 • 857.75/88.471
= 9.695

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
= 1899.5/ (2 • 1400.0) • 392.887 • 857.75/2
= 114.312 Kg-m.

Compressive Allowable, KL/r < Cc (9.6952 < 125.9015) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (9.7)^2 / (2 \cdot 125.9^2)) 249 / (5/3 + 3 \cdot (9.7) / (8 \cdot 125.9) - (9.7^3) / (8 \cdot 125.9^3))$
= 146.4 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba
= 1.88/146.43 + (114.31 • 187.5/43945308)/166.0
= 0.042

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	392.9	10.0	0.0	39.3	0.0	3.27
Totals	...	...	...	75.8	...	0.440E+04

Distance to Centroid from Datum [ytot]:

= AY / A
= 0.0/75.789
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2



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$$= 375.0/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1220.0 - \sqrt{(680.5 + 14.0)^2 - (392.887/2)^2} - 22.0$$

$$= 546.471 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4397.7/75.789)}$$

$$= 76.175 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= \text{KL}/r$$

$$= 1 \cdot 546.471/76.175$$

$$= 7.174$$

Unit Force [Force,u]:

$$= \text{FI} / (2 \cdot \text{Baseplate Length})$$

$$= 1899.514 / (2 \cdot 1400.0)$$

$$= 0.678 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.678 \cdot 392.887 \cdot 546.471$$

$$= 145.656 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 145.656/234545.141$$

$$= 6.090 \text{ N./mm}^2$$

Compressive Allowable, $\text{KL}/r < C_c$ (7.1739 < 125.9015) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (\text{KL}/r) / (8 \cdot C_c) - (\text{KL}/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.17)^2 / (2 \cdot 125.9^2)) 249 / (5/3 + 3 \cdot (7.17) / (8 \cdot 125.9) - (7.17^3) / (8 \cdot 125.9^3))$$

$$= 147.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= \text{Sc}/\text{Sca} + (\text{Mbase,c} \cdot \text{C1/I}) / \text{Sba}$$

$$= 2.79/147.27 + (145.66 \cdot 187.5/4397.717) / 166.0$$

$$= 0.056$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	148.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load Q0 (Weight)	Q0	4740.3 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	585.2 Kg
Maximum Transverse Force	Ft	949.8 Kg
Maximum Longitudinal Force	F1	1899.5 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 8781.3 \text{ Kg}$$



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Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= \max(F_f; F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 8781.3 / (3.043 \cdot 6)$$

$$= 47.2 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 949.8 / (3.043 \cdot 6)$$

$$= 5.1 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 1899.51 / (82.8 \cdot 6.0)$$

$$= 0.3750 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1220.0 \cdot 949.76 + 0.0$$

$$= 1158.73 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 1158.73 / 4740.28$$

$$= 244.44 \text{ mm} > B_{plen} / 6 \text{ --> Uplift in Transverse direction}$$

f = Bplen / 2 - Edgedis

$$= 1400.0 / 2 - 50.8$$

$$= 649.20 \text{ mm}$$

Modular Ratio Of Steel/Concrete (n1):

$$= E_S / E_C$$

$$= 203402.5 / 21526.32$$

$$= 9.45$$

K1 = 3 (e - 0.5 • Bplen)

$$= 3 (244.44 - 0.5 \cdot 1400.0)$$

$$= -1366.69 \text{ mm}$$

K2 = 6 • n1 • At / Bpwid • (f + e)

$$= 6 \cdot 9.45 \cdot 9.13 / 400.0 \cdot (649.2 + 244.44)$$

$$= 115619.32 \text{ mm}^2$$

K3 = -K2 • (0.5 • Bplen + f)

$$= -115619.32 \cdot (0.5 \cdot 1400.0 + 649.2)$$

$$= -155993580.00 \text{ mm}^3$$

Iteratively Solving for the Effective Bearing Length:

$$Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$$

$$Y^3 + -1366.69 \cdot Y^2 + 115619.32 \cdot Y + -0.2E+09 = 0$$

$$Y = 1365.67 \text{ mm}$$

Num = (Bplen / 2 - Y / 3 - e)

$$= (1400.0 / 2 - 1365.67 / 3 - 244.44)$$

$$= 0.34$$

Denom = (Bplen / 2 - Y / 3 + f)

$$= (1400.0 / 2 - 1365.67 / 3 + 649.2)$$

$$= 893.98$$



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Total Bolt Tension Force [Tforce]:

$$= - QO \cdot Num / Denom$$

$$= - 4740.28 \cdot 0.34 / 893.98$$

$$= -1.80 \text{ Kgf}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$= Tforce / (Stba \cdot Nbt)$$

$$= -1.8 / (138.0 \cdot 3.0)$$

$$= -0.0004 \text{ cm}^2$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.375, -0.0004]$$

$$= 0.3750 \text{ cm}^2$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$= 2(QO + Tforce) / (Y \cdot Bpwid)$$

$$= 2(4740.28 + -1.8) / (1365.67 \cdot 400.0)$$

$$= 170.12 \text{ KPa.}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$= (Bplen - WebIngh) / 2$$

$$= (1400.0 - 1349.2) / 2$$

$$= 25.4000 \text{ mm}$$

Overturning Moment due To Bolt Tension [Mt]:

$$= Tforce \cdot Adist$$

$$= -1.8 \cdot 25.4$$

$$= -0.05 \text{ Kg-m.}$$

Equivalent Bearing Pressure (f1):

$$= fc \cdot (Y - Adist) / Y$$

$$= 170.12 \cdot (1365.67 - 25.4) / 1365.67$$

$$= 166.96 \text{ KPa.}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc)$$

$$= (25.4^2 \cdot 400.0 / 6) \cdot (166.96 + 2 \cdot 170.12)$$

$$= 2.22 \text{ Kg-m.}$$

Baseplate Required Thickness [Treq]:

$$= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$$

$$= \sqrt{(6 \cdot \max(-0.05, 2.22) / (400.0 \cdot 222.0))}$$

$$= 1.2141 \text{ mm}$$

MDMT Calculations for: RH saddle fixed, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 14.0, c = 3.0 mm, E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.971, Temp. Reduction = 2 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -20 °C

Review notes about nozzle loadings and supports in the Warnings report.

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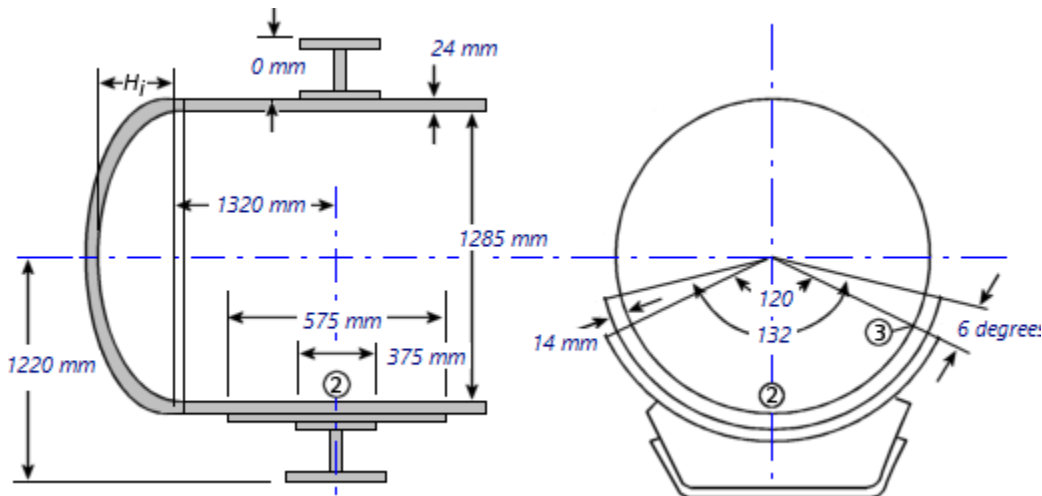
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	656.00	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	6650.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1320.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	575.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	148.00	N./mm ²
Shell Allowable Stress used in Calculation		278.99	N./mm ²
Head Allowable Stress used in Calculation		278.99	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1220.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	27672.04	Kgf
Pressure used in Analysis	P	5661.430 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	84.85	278.99
Long. Stress at Bottom of Midspan	92.02	278.99
Long. Stress at Top of Saddles	109.02	278.99
Long. Stress at Bottom of Saddles	77.02	278.99
Tangential Shear in Shell	13.91	165.23
Circ. Stress at Horn of Saddle	21.04	418.49
Circ. Compressive Stress in Shell	1.06	278.99



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(334.9/2 + 0) \cdot 1220.0/1178.6606$$

$$= 519.9 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(129.58, 0) \cdot 1220.0/3960.001$$

$$= 39.9 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 27152 + \max(40, 520, 0, 0)$$

$$= 27672.0 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(0.0 + \pi/4(1.633 \cdot 1.2)^2)$$

$$= 2541.528 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	28018.80	Kgf
Transverse Shear Load	167.43	Kgf
Longitudinal Shear Load	129.58	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1285.0/2 + 24.0 + 14.0)\sin(120.0/2), 2 \cdot 656.0)$$

$$= 1178.661 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -27672 \cdot 1320.0 [1 - (1 - 1320.0/6650.8 + (656.0^2 - 0.0^2)/(2 \cdot 1320.0 \cdot 6650.8))/(1 + (4 \cdot 0.0)/(3 \cdot 6650.8))]$$

$$= -6354.5 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 27672 \cdot 6651/4((1 + 2(656^2 - 0^2)/(6651^2))/(1 + (4 \cdot 0)/(3 \cdot 6651)) - 4 \cdot 1320/6651)$$

$$= 10378.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 5661.43 \cdot 656.0/(2 \cdot 21.0) - 10378.7/(\pi \cdot 656.0^2 \cdot 21.0)$$

$$= 84.85 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$



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$$= 5661.43 \cdot 656.0 / (2 \cdot 21.0) + 10378.7 / (\pi \cdot 656.0^2 \cdot 21.0)$$

$$= 92.02 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 5661.43 \cdot 656.0 / (2 \cdot 21.0) - 6354.5 / (0.1066 \cdot \pi \cdot 656.0^2 \cdot 21.0)$$

$$= 109.02 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 5661.43 \cdot 656.0 / (2 \cdot 21.0) + 6354.5 / (0.1923 \cdot \pi \cdot 656.0^2 \cdot 21.0)$$

$$= 77.02 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 27672(6650.8 - 2 \cdot 1320.0) / (6650.8 + (4 \cdot 0.0 / 3))$$

$$= 16687.8 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 16687.77 / (656.0 \cdot 21.0)$$

$$= 13.91 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(656.0 \cdot 21.0)}$$

$$= 91.550 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(375.0 + 1.56 \cdot \sqrt{(656.0 \cdot 21.0)}, 2 \cdot 1320.0)$$

$$= 558.10 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 27672 \cdot 0.1 / (558.099(21.0 + 1.0 \cdot 14.0))$$

$$= -1.06 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -27672 / (4(21.0 + 1.0 \cdot 14.0)558.099) -$$

$$3 \cdot 0.053 \cdot 27672 / (2(21.0 + 1.0 \cdot 14.0)^2)$$

$$= -21.04 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1400.0000	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	N./mm ²
Height of Web at Center	Hw,c	560.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(645.5 \cdot 21.0)} + 575.0)^3) / 12$$

$$= 58.393 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

= ShellInertia + AShell(C1 - Y)²

= 58.393 + 158.892(181.23 - 10.5)²

= 46373.379 cm⁴

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	756.6	21.0	10.5	158.9	166836.4	0.464E+05
Wearplate	575.0	14.0	28.0	80.5	225400.0	0.189E+05
Web	10.0	517.5	293.8	51.8	1520156.1	0.181E+05
BasePlate	400.0	22.0	563.5	88.0	4958799.5	0.129E+06
Totals	...	...	...	379.1	6871192.0	0.212E+06

Distance to Centroid [C1]:

= AY / A

= 6871192/379.142

= 181.230 mm

Angle [β]:

= 180 - Saddle Angle/2

= 180 - 120.0/2

= 120.0

Saddle Splitting Coefficient [K1]:

= (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))

= (1 + cos(120.0) - 0.5•sin(120.0)²)/(π - 2.094 + sin(120.0)cos(120.0))

= 0.2035

Saddle Splitting Force [Fh]:

= K1 • Q

= 0.204 • 27672.035

= 5631.8594 Kgf

Tension Stress, St = (Fh/As) = 2.5076 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 185.9966 N./mm²

Saddle Splitting Dimension [d]:

= B - R • sin(theta/2)/(theta/2 in radians)

= 1220.0 - 645.5 • sin(120.0/2)/1.0472

= 686.176 mm

Bending Moment, M = Fh * d = 3864.5220 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 3.2395 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 206.6629 N./mm²

Minimum Thickness of Baseplate per Moss:

= √(3(Q + SaddleWt)BasePlateWidth / (4 • BasePlateLength • AllStress))

= √(3(27672 + 347)400.0/(4 • 1400.0 • 206.663))

= 16.879 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)

= 2 • cos(90 - 120.0/2)(642.5 + 24.0 + 14.0)

= 1178.661 mm

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= 1178.6606/(4 - 1)

= 392.887 mm



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 392.8869 \cdot 400.0 / 2$$

$$= 785.774 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 28018.797 / (1400.0 \cdot 400.0)$$

$$= 5.003 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 785.8 \cdot 5.0$$

$$= 3931.506 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 36.5 + 19.644$$

$$= 56.144 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 3931.5 / 56.1443$$

$$= 6.867 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 375.0 - 10.0$$

$$= 365.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 56.144$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 375.0 / 2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (4394.5 / 56.144) }$$

$$= 88.471 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1220.0 - \cos(120.0 / 2) (642.5 + 24.0 + 14.0) - 22.0$$

$$= 857.750 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 0.19994E+09 / 310.0) }$$

$$= 112.838$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 857.75/88.471$$

$$= 9.695$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 129.6 / (2 \cdot 1400.0) \cdot 392.887 \cdot 857.75/2$$

$$= 7.798 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ (9.6952 < 112.8376) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (9.7)^2 / (2 \cdot 112.84^2)) 310 /$$

$$(5/3 + 3 \cdot (9.7) / (8 \cdot 112.84) - (9.7^3) / (8 \cdot 112.84^3))$$

$$= 181.8 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 6.87/181.8 + (7.8 \cdot 187.5/43945308) / 206.66$$

$$= 0.039$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	392.9	10.0	0.0	39.3	0.0	3.27
Totals	...	...	...	75.8	...	0.440E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/75.789$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375.0/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1220.0 - \sqrt{ (680.5 + 14.0)^2 - (392.887/2)^2 } - 22.0$$

$$= 546.471 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (4397.7/75.789) }$$

$$= 76.175 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 546.471/76.175$$

$$= 7.174$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 129.584 / (2 \cdot 1400.0)$$

$$= 0.046 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.046 \cdot 392.887 \cdot 546.471$$

$$= 9.937 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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$$= 9.937/234545.141$$

$$= 0.415 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.1739 < 112.8376$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r)^2 / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.17)^2 / (2 \cdot 112.84^2)) 310 / (5/3 + 3 \cdot (7.17) / (8 \cdot 112.84) - (7.17^3) / (8 \cdot 112.84^3))$$

$$= 183 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 10.18 / 183.01 + (9.94 \cdot 187.5 / 4397.717) / 206.66$$

$$= 0.058$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	148.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load QO (Weight)	QO	27498.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	39.9 Kg
Maximum Transverse Force	Ft	167.4 Kg
Maximum Longitudinal Force	F1	129.6 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 129.6 / (3.043 \cdot 6)$$

$$= 0.7 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 167.4 / (3.043 \cdot 6)$$

$$= 0.9 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 129.58 / (28.98 \cdot 6.0)$$

$$= 0.0731 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1220.0 \cdot 167.43 + 0.0$$

$$= 204.27 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 204.27 / 27498.89$$

$$= 7.43 \text{ mm} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$



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Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0.0, 0.0731, 0.0]

= 0.0731 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	656.00	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	6650.80	mm
Distance from Saddle to Vessel tangent	a1 or a	1320.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	575.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	148.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	334.25	mm
Shell Allowable Stress used in Calculation		278.99	N./mm ²
Head Allowable Stress used in Calculation		278.99	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1220.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		4035.95	Kgf
Pressure used in Analysis	P	5661.430	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	88.07	278.99
Long. Stress at Bottom of Midspan	88.79	278.99
Long. Stress at Top of Saddles	92.43	278.99
Long. Stress at Bottom of Saddles	86.21	278.99
Tangential Shear in Shell	1.90	165.23
Circ. Stress at Horn of Saddle	3.07	418.49
Circ. Compressive Stress in Shell	0.15	278.99

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3.0(334.9/2 + 0) • 1220.0/1178.6606

= 519.9 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= max(Fl,wind; Sum of X Forces) • B / Ls

= max(129.58, 0) • 1220.0/3960.001

= 39.9 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:



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$$\begin{aligned}
 &= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst}) \\
 &= 3516 + \max(40, 520, 0, 0) \\
 &= 4035.9 \text{ Kg}
 \end{aligned}$$

Longitudinal Wind Force [F_L]:

$$\begin{aligned}
 &= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2)) \\
 &= 1.1 \cdot 766.08(0.0 + \pi/4(1.633 \cdot 1.2)^2) \\
 &= 2541.528 \text{ N}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	4382.71	Kgf
Transverse Shear Load	167.43	Kgf
Longitudinal Shear Load	129.58	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m) \\
 &= \min(2(1285.0/2 + 24.0 + 14.0)\sin(120.0/2), 2 \cdot 656.0) \\
 &= 1178.661 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to R_m/2.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)] \\
 &= -4036 \cdot 1320.0 [1 - (1 - 1320.0/6650.8 + (656.0^2 - 334.25^2)/(2 \cdot 1320.0 \cdot 6650.8))/(1 + (4 \cdot 334.25)/(3 \cdot 6650.8))] \\
 &= -1234.9 \text{ Kg-m}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L) \\
 &= 4036 \cdot 6651/4 ((1 + 2(656^2 - 334^2)/(6651^2))/(1 + (4 \cdot 334)/(3 \cdot 6651)) - 4 \cdot 1320/6651) \\
 &= 1052.3 \text{ Kg-m}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ₁]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t) \\
 &= 5661.43 \cdot 656.0/(2 \cdot 21.0) - 1052.3/(\pi \cdot 656.0^2 \cdot 21.0) \\
 &= 88.07 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ₂]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t) \\
 &= 5661.43 \cdot 656.0/(2 \cdot 21.0) + 1052.3/(\pi \cdot 656.0^2 \cdot 21.0) \\
 &= 88.79 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*₃]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t) \\
 &= 5661.43 \cdot 656.0/(2 \cdot 21.0) - 1234.9/(0.1066 \cdot \pi \cdot 656.0^2 \cdot 21.0) \\
 &= 92.43 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*₄]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M1/(K1* \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 5661.43 \cdot 656.0/(2 \cdot 21.0) + 1234.9/(0.1923 \cdot \pi \cdot 656.0^2 \cdot 21.0) \\
 &= 86.21 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$



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$$= 4036(6650.8 - 2 \cdot 1320.0) / (6650.8 + (4 \cdot 334.25/3))$$

$$= 2281.0 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 2281.05 / (656.0 \cdot 21.0)$$

$$= 1.90 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(656.0 \cdot 21.0)}$$

$$= 91.550 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(375.0 + 1.56 \cdot \sqrt{(656.0 \cdot 21.0)}, 2 \cdot 1320.0)$$

$$= 558.10 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 4036 \cdot 0.1 / (558.099(21.0 + 1.0 \cdot 14.0))$$

$$= -0.15 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -4036 / (4(21.0 + 1.0 \cdot 14.0)558.099) -$$

$$3 \cdot 0.053 \cdot 4036 / (2(21.0 + 1.0 \cdot 14.0)^2)$$

$$= -3.07 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1400.0000	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	N./mm ²
Height of Web at Center	Hw, c	560.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(645.5 \cdot 21.0)} + 575.0)21.0^3) / 12$$

$$= 58.393 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 58.393 + 158.892(181.23 - 10.5)^2$$

$$= 46373.379 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	756.6	21.0	10.5	158.9	166836.4	0.464E+05
Wearplate	575.0	14.0	28.0	80.5	225400.0	0.189E+05
Web	10.0	517.5	293.8	51.8	1520156.1	0.181E+05
BasePlate	400.0	22.0	563.5	88.0	4958799.5	0.129E+06
Totals	...	...	...	379.1	6871192.0	0.212E+06



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Distance to Centroid [C1]:

$$= AY / A$$

$$= 6871192/379.142$$

$$= 181.230 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 4035.947$$

$$= 821.4026 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.3657 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 185.9966 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1220.0 - 645.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 686.176 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 563.6378 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.4725 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 206.6629 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(4036 + 347) \cdot 400.0) / (4 \cdot 1400.0 \cdot 206.663)}$$

$$= 6.676 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (642.5 + 24.0 + 14.0)$$

$$= 1178.661 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 1178.6606 / (4 - 1)$$

$$= 392.887 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 392.8869 \cdot 400.0/2$$

$$= 785.774 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 4382.71 / (1400.0 \cdot 400.0)$$

$$= 0.783 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 785.8 \cdot 0.78$$

$$= 614.968 \text{ Kgf}$$



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Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= 36.5 + 19.644

= 56.144 cm²

Compressive Stress [Sc]:

= P/Ar

= 615.0/56.1443

= 1.074 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 375.0 - 10.0

= 365.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/56.144

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 375.0/2

= 187.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{4394.5/56.144}$

= 88.471 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 1220.0 - cos(120.0/2)(642.5 + 24.0 + 14.0) - 22.0

= 857.750 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/310.0}$

= 112.838

Slenderness ratio [KL/r]:

= KL/r

= 1 • 857.75/88.471

= 9.695

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 129.6 / (2 • 1400.0) • 392.887 • 857.75/2

= 7.798 Kg-m.

Compressive Allowable, KL/r < Cc (9.6952 < 112.8376) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (9.7)^2 / (2 \cdot 112.84^2)) 310 /$

$(5/3 + 3 \cdot (9.7) / (8 \cdot 112.84) - (9.7^3) / (8 \cdot 112.84^3))$

= 181.8 N./mm²



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AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 1.07/181.8 + (7.8 \cdot 187.5/43945308) / 206.66$$

$$= 0.007$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	392.9	10.0	0.0	39.3	0.0	3.27
Totals	...	...	...	75.8	...	0.440E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/75.789$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375.0/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1220.0 - \sqrt{ (680.5 + 14.0)^2 - (392.887/2)^2 } - 22.0$$

$$= 546.471 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 4397.7/75.789 }$$

$$= 76.175 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 546.471/76.175$$

$$= 7.174$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 129.584/(2 \cdot 1400.0)$$

$$= 0.046 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.046 \cdot 392.887 \cdot 546.471$$

$$= 9.937 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 9.937/234545.141$$

$$= 0.0415 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ (7.1739 < 112.8376) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (7.17)^2 / (2 \cdot 112.84^2)) 310 / (5/3 + 3 \cdot (7.17) / (8 \cdot 112.84) - (7.17^3) / (8 \cdot 112.84^3))$$

$$= 183 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 1.59/183.01 + (9.94 \cdot 187.5/4397.717) / 206.66$$

$$= 0.011$$



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Saddle Calcs: Test Case:

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Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Material Specification	SA-307 A	
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400 cm ²
BasePlate Allowable Stress	S	148.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.0428 cm ²
Saddle Load Q0 (Weight)	Q0	3862.8 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	39.9 Kgf
Maximum Transverse Force	Ft	167.4 Kgf
Maximum Longitudinal Force	Fl	129.6 Kgf
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 129.6 / (3.043 \cdot 6)$$

$$= 0.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 167.4 / (3.043 \cdot 6)$$

$$= 0.9 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 129.58 / (82.8 \cdot 6.0)$$

$$= 0.0256 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1220.0 \cdot 167.43 + 0.0$$

$$= 204.27 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 204.27 / 3862.81$$

$$= 52.88 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$$

$$= \max[0.0, 0.0256, 0.0]$$

$$= 0.0256 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N1-Tube Inlet

Nozl: 11

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Input, Nozzle Desc: N1-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	5900.000	KPa.
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1285.00	mm
Design Length of Section	L	999.9695	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1203.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		100.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-336 F11	
Hub Height of Integral Nozzle	h	26.1658	mm
Height of Beveled Transition	L'	16.1658	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		40.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



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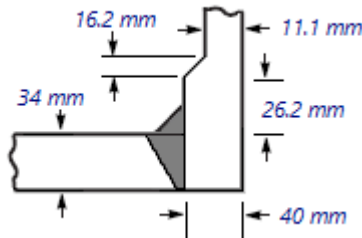
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Nozzle Calcs.: N1-Tube Inlet

Nozl: 11

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Reinforcement CALCULATION, Description: N1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	92.050 mm.
Actual Thickness Used in Calculation	11.125 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5900.0 \cdot 645.5 / (147 \cdot 1.0 - 0.6 \cdot 5900.0)$
 $= 26.5487 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5900.0 \cdot 49.02 / (147 \cdot 1.0 - 0.6 \cdot 5900.0)$
 $= 2.0163 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5135 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 28.875 / 0.5773$
 $= 50.0126 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	196.0992 mm
Parallel to Vessel Wall, opening length	d	98.0496 mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	62.6446 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 147.0 / 147.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 147.0 / 147.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$



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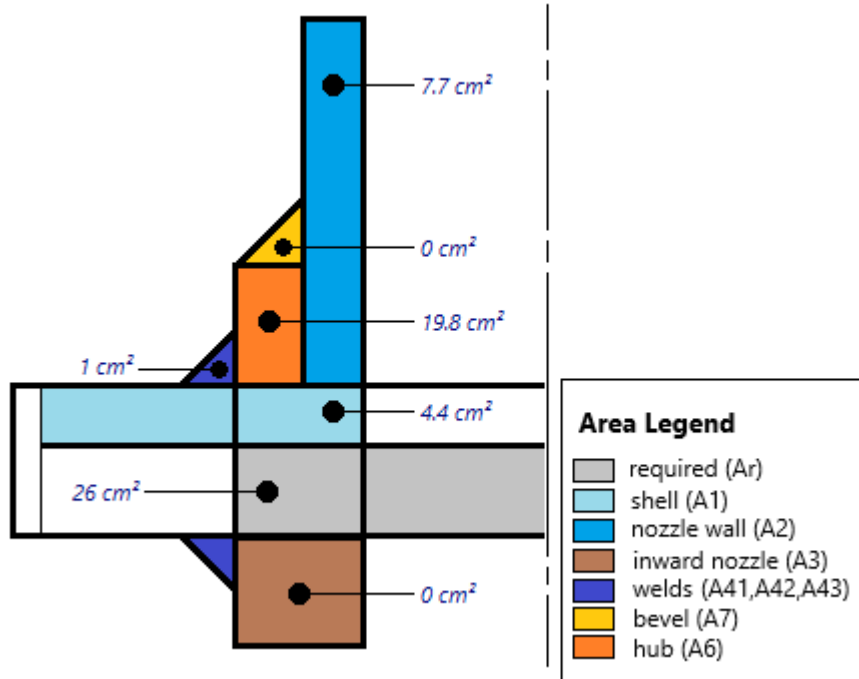
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= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	26.031	1.893	NA
Area in Shell	A1	4.364	26.609	NA
Area in Nozzle Wall	A2	7.654	9.537	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	19.778	19.778	NA
TOTAL AREA AVAILABLE	Atot	32.797	56.924	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (98.0496 \cdot 26.5487 \cdot 1.0 + 2 \cdot 37.0 \cdot 26.5487 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 26.031 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 98.05(1.0 \cdot 31.0 - 1.0 \cdot 26.549) - 2 \cdot 37.0$$

$$(1.0 \cdot 31.0 - 1.0 \cdot 26.5487) \cdot (1 - 1.0)$$

$$= 4.364 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn) fr2$$



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$$= (2 \cdot 62.64)(8.13 - 2.02)1.0$$

$$= 7.654 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(62.6, 250.0, 26.2)) \cdot (40.0 - 11.1) \cdot 1.0$$

$$= 19.778 \text{ cm}^2 \text{ Includes Bevel Area of } 4.668 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.0163 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 29.5487 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 29.5487 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.4543 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.2578 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.258, \max(29.5487, 4.5)]$$

$$= 8.2578 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.0163, 8.2578)$$

$$= 8.2578 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 11.125 = 9.735 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 55.2,	Allowable	: 147.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 313.6 N./mm ²	Passed
Occasional	: 16.4,	Allowable	: 195.5 N./mm ²	Passed
Shear	: 26.1,	Allowable	: 102.9 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 9.735, tr = 2.002, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.297$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 34.0, tr = 26.365, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-1 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -1 °C



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Weld Size Calculations, Description: N1-Tube Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (26.0309 - 4.3645 + 2 \cdot 37.0 \cdot 1.0 \cdot (1.0 \cdot 31.0 - 26.5487)) \cdot 147)$
 $= 37414.49 \text{ Kg}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (27.4323 + 1. - 0.0 \cdot 1.0) \cdot 147$
 $= 42618.76 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (27.4323 + 0.0 + 1. + (22.94)) \cdot 147$
 $= 77004.80 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (27.4323 + 0.0 + 1. + 0.0 + (22.94)) \cdot 147$
 $= 77004.80 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 172.0496 \cdot 10.0 \cdot 0.49 \cdot 147$
 $= 19850. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 67.5248) \cdot (40.0 - 3.0) \cdot 0.7 \cdot 147$
 $= 82360. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 172.0496 \cdot (30.0 - 0.0) \cdot 0.74 \cdot 147$
 $= 89935. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (19850 + 82360) = 102210 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (19850 + 0 + 89935 + 0) = 109785 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$



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$$= (19850 + 89935 + 0) = 109785 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 102209 Kgf, must exceed W = 37414 Kgf or W1 = 42618 Kgf

Path 2-2 = 109784 Kgf, must exceed W = 37414 Kgf or W2 = 77004 Kgf

Path 3-3 = 109784 Kgf, must exceed W = 37414 Kgf or W3 = 77004 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6611 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 5.7850 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 289.7850 mm

Input Echo, WRC297 Item 1, Description: N1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1285.000	mm
Shell Thickness	34.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 148.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 147.00	N./mm ²

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	92.050	mm
Nozzle Thickness	40.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 148.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 147.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	5900.00	KPa.
Radial Load	P	-917.81	Kgf
Circumferential Shear	Vc	662.86	Kgf
Longitudinal Shear	Vl	662.86	Kgf
Circumferential Moment	Mc	203.88	Kg-m.
Longitudinal Moment	Ml	203.88	Kg-m.
Torsional Moment	Mt	234.46	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (4.301) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 172.05 + 2 \cdot 1.65 \cdot \sqrt{(661.0(34.0 - 3.0))}$$

$$= 644.434 \text{ mm}$$



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WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1322.000 mm
 Nozzle Outside Diameter (d) = 172.050 mm
 Vessel Thickness used (T) = 31.000 mm
 Nozzle Thickness used (t) = 37.000 mm
 T / t = 0.838
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.850$$

Nr/P = 0.131
 Mr/P = 0.221
 M0/P = 0.072
 N0/P = 0.134
 MrD/Mc = 0.292
 NrDL/Mc = 0.127
 M0d/Mc = 0.106
 N0DL/Mc = -0.031
 MrD/Ml = 0.215
 NrDL/Ml = 0.111
 M0D/Ml = 0.079
 N0DL/Ml = 0.098

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-5	5	5	-5
Normal Pressure Stress	119	125	119	125
Outplane Stress Summary	118	127	132	118

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-15	15	15	-15
Inplane Pressure Stress	59	59	59	59
Inplane Stress Summary	56	63	90	34

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0



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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress	118	127	132	118
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-7	7	7	-7
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	59	59	59	59

Outplane Stress Summary	57	65	72	49
-------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-21	21	21	-21
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	119	125	119	125

Inplane Stress Summary	110	134	156	94
------------------------	-----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	110	134	156	95
------------------------	-----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1



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Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	13	6	13
Hoop Stress Summary	6	13	9	16

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-4	-4	4	4
Axial Bending (ML)	-8	8	8	-8
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-2	1	22	-5

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	10	14	22	22

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	13	6	13
Hoop Stress Summary	8	15	7	14

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	-4	-4	4	4
Axial Bending (Mc)	-11	11	11	-11
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-5	4	26	-9



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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	14	15	26	24

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	119	125	119	125	119	125	119	125
Circ. Pl (SUS)	0	0	2	2	0	0	2	2
Circ. Q (SUS)	-1	1	9	-9	-9	9	33	-33
Long. Pm (SUS)	59	59	59	59	59	59	59	59
Long. Pl (SUS)	0	0	2	2	1	1	1	1
Long. Q (SUS)	-3	3	27	-27	-3	3	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	119.0	125.0	119.0	125.0	119.0	125.0	119.0	125.0
Pm+Pl (SUS)	119.0	125.0	121.0	127.0	119.0	125.0	121.0	127.0
Pm+Pl+Q (Total)	118.0	126.0	130.0	118.0	110.0	134.0	154.0	94.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	125.00	147.00	Passed
Pm+Pl (SUS)	127.00	220.50	Passed
Pm+Pl+Q (TOTAL)	154.01	441.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	13	6	13	6	13	6	13
Circ. Pl (SUS)	0	0	2	2	1	1	1	1



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Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-4	-4	4	4	-4	-4	4	4
Long. Q (SUS)	-2	2	14	-14	-5	5	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	6.0	13.0	6.0	13.0	6.0	13.0	6.0	13.0
Pm+Pl (SUS)	7.0	14.0	8.0	15.0	8.0	15.0	7.0	14.0
Pm+Pl+Q (Total)	9.2	13.1	21.1	22.1	13.2	14.1	24.1	24.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	13.00	147.00	Passed
Pm+Pl (SUS)	15.00	220.51	Passed
Pm+Pl+Q (TOTAL)	24.08	441.01	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N2-Tube Outlet

Nozl: 12

9:46am

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Input, Nozzle Desc: N2-Tube Outlet

From: 30

Pressure for Reinforcement Calculations	P	5912.596	KPa.
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1285.00	mm
Design Length of Section	L	999.9695	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1003.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		80.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-336 F11	
Hub Height of Integral Nozzle	h	26.9060	mm
Height of Beveled Transition	L'	23.0940	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



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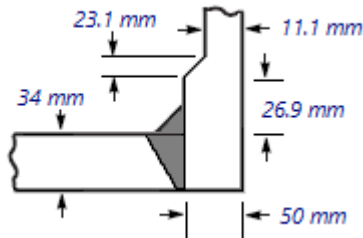
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Nozzle Calcs.: N2-Tube Outlet

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Reinforcement CALCULATION, Description: N2-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	66.650 mm.
Actual Thickness Used in Calculation	11.125 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $5912.6 \cdot 645.5 / (147 \cdot 1.0 - 0.6 \cdot 5912.6)$

= 26.6068 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $5912.6 \cdot 36.32 / (147 \cdot 1.0 - 0.6 \cdot 5912.6)$

= 1.4972 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4458 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

= (Hub Thickness - Neck Thickness) / $\tan(30)$

= $31.0 / 0.5773$

= 53.6936 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	150.9000 mm
--	----	-------------

Parallel to Vessel Wall	Rn+tn+t	75.4500 mm
-------------------------	---------	------------

Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	70.3130 mm
---	------	------------

Weld Strength Reduction Factor [fr1]:

= $\min(1, S_n / S_v)$

= $\min(1, 147.0 / 147.0)$

= 1.000

Weld Strength Reduction Factor [fr2]:

= $\min(1, S_n / S_v)$

= $\min(1, 147.0 / 147.0)$

= 1.000



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Nozzle Calcs.: N2-Tube Outlet

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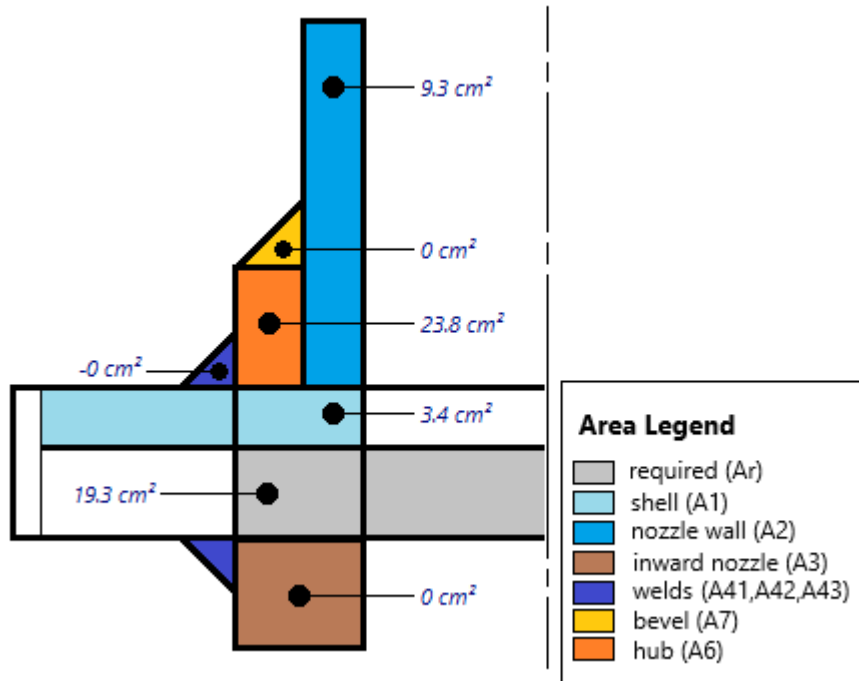
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Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(1.0, 1.0)

= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	19.330	1.403	NA
Area in Shell	A1	3.438	21.236	NA
Area in Nozzle Wall	A2	9.321	10.799	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	-0.000	-0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	23.841	23.841	NA
TOTAL AREA AVAILABLE	Atot	36.599	55.876	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)

= (72.6496•26.6068•1.0+2•39.1252•26.6068•1.0•(1-1.0))

= 19.330 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)

= 78.25(1.0 • 31.0 - 1.0 • 26.607) - 2 • 39.125

(1.0 • 31.0 - 1.0 • 26.6068) • (1 - 1.0)

= 3.438 cm²



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Nozzle Calcs.: N2-Tube Outlet

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Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 70.31) (8.13 - 1.5) 1.0$$

$$= 9.321 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= (W_o^2 - \text{Area Lost}) \cdot f_{r2} + (W_i - \text{can}/0.707)^2 - \text{Area Lost} \cdot f_{r2}$$

$$= (10.0^2 - 1.0) \cdot 1.0 + (0.0^2 - 0.0) \cdot 1.0$$

$$= -0.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(70.3, 250.0, 26.9)) \cdot (42.1 - 11.1) \cdot 1.0$$

$$= 23.841 \text{ cm}^2 \text{ Includes Bevel Area of } 7.159 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness per Internal/External pressures	ta = 4.4972 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 29.6068 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 29.6068 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.4543 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.8, \max(29.6068, 4.5)]$$

$$= 7.8000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.4972, 7.8)$$

$$= 7.8000 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 11.125 = 9.735 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 55.8,	Allowable	: 147.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 313.0 N./mm ²	Passed
Occasional	: 11.9,	Allowable	: 195.5 N./mm ²	Passed
Shear	: 29.8,	Allowable	: 102.9 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 9.735, tr = 1.487, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.221$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 34.0, tr = 26.423, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.852$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-1 °C



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Nozzle Calcs.: N2-Tube Outlet Nozl: 12 9:46am Sep 9, 2026

Gov. MDMT of the nozzle to shell joint welded assembly : -1 °C

Weld Size Calculations, Description: N2-Tube Outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (19.3297 - 3.4377 + 2 \cdot 39.1252 \cdot 1.0 \cdot (1.0 \cdot 31.0 - 26.6068)) \cdot 147) \\
 &= 28974.38 \text{ Kgf}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_r2) \cdot S_v \\
 &= (33.1615 + -0.0 - 0.0 \cdot 1.0) \cdot 147 \\
 &= 49707.64 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v \\
 &= (33.1615 + 0.0 + -0.0 + (24.2576)) \cdot 147 \\
 &= 86068.74 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S \\
 &= (33.1615 + 0.0 + -0.0 + 0.0 + (24.2576)) \cdot 147 \\
 &= 86068.74 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 150.9 \cdot 10.0 \cdot 0.49 \cdot 147 \\
 &= 17410. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 55.8874) \cdot (42.1252 - 3.0) \cdot 0.7 \cdot 147 \\
 &= 72081. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 150.9 \cdot (30.0 - 0.0) \cdot 0.74 \cdot 147 \\
 &= 78879. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (17410 + 72081) = 89491 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (17410 + 0 + 78879 + 0) = 96289 \text{ Kgf}
 \end{aligned}$$



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Nozzle Calcs.: N2-Tube Outlet

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Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (17410 + 78879 + 0) = 96289 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 89490 Kgf, must exceed W = 28974 Kgf or W1 = 49707 Kgf

Path 2-2 = 96289 Kgf, must exceed W = 28974 Kgf or W2 = 86068 Kgf

Path 3-3 = 96289 Kgf, must exceed W = 28974 Kgf or W3 = 86068 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6849 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 4.4455 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 288.4455 mm

Input Echo, WRC297 Item 1, Description: N2-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1285.000	mm	
Shell Thickness	34.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-387 11		
Vessel Cold S.I. Allowable	Smc 148.00	N./mm ²	
Vessel Hot S.I. Allowable	Smh 147.00	N./mm ²	
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	66.650	mm	
Nozzle Thickness	50.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-336 F11		
Nozzle Cold S.I. Allowable	SNmc 148.00	N./mm ²	
Nozzle Hot S.I. Allowable	SNmh 147.00	N./mm ²	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	5912.60	KPa.
Radial Load	P	-734.25	Kgf
Circumferential Shear	Vc	530.29	Kgf
Longitudinal Shear	Vl	530.29	Kgf
Circumferential Moment	Mc	130.48	Kg-m.
Longitudinal Moment	Ml	130.48	Kg-m.
Torsional Moment	Mt	150.05	Kg-m.

Include Axial Pressure Thrust No

Include Pressure Stress Indices per Div. 2 No

Local Loads applied at end of Nozzle/Attachment No

Warning - The ratio, Dn/Tv, (4.901) is < 5
check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (3.333) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:



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Nozzle Calcs.: N2-Tube Outlet

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Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 166.65 + 2 \cdot 1.65 \cdot \sqrt{661.0(34.0 - 3.0)}$$

$$= 639.034 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1322.000 mm

Nozzle Outside Diameter (d) = 166.650 mm

Vessel Thickness used (T) = 31.000 mm

Nozzle Thickness used (t) = 47.000 mm

T / t = 0.660

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.823$$

Nr/P = 0.123

Mr/P = 0.239

M0/P = 0.073

N0/P = 0.140

MrD/Mc = 0.301

NrDL/Mc = 0.106

M0d/Mc = 0.102

N0DL/Mc = -0.001

MrD/Ml = 0.219

NrDL/Ml = 0.100

M0D/Ml = 0.076

N0DL/Ml = 0.099

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-3	3	3	-3
Normal Pressure Stress	120	126	120	126
Outplane Stress Summary	120	126	128	121

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	10	-10	10	-10
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-10	10	10	-10
Inplane Pressure Stress	60	60	60	60
Inplane Stress Summary	60	59	83	40

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------



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Nozzle Calcs.: N2-Tube Outlet Nozl: 12 9:46am Sep 9,2026

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	0	0
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	120	126	128	121
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-4	4	4	-4
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	60	60	60	60

Outplane Stress Summary	59	62	69	52
-------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	10	-10	10	-10
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-14	14	14	-14
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	120	126	120	126

Inplane Stress Summary	116	129	147	102
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	1	1
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	116	129	147	102
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside



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Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	10	3	10

Hoop Stress Summary	3	10	5	12
---------------------	---	----	---	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	0	0	11	-2
----------------------	---	---	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	1	1	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	5	11	11	14
------------------------	---	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	10	3	10

Hoop Stress Summary	4	11	4	11
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	-2	-2	2	2



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Nozzle Calcs.: N2-Tube Outlet Nozl: 12 9:46am Sep 9,2026

Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-2	0	13	-3
----------------------	----	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	6	11	13	15

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	120	126	120	126	120	126	120	126
Circ. Pl (SUS)	1	1	1	1	0	0	0	0
Circ. Q (SUS)	0	0	6	-6	-4	4	24	-24
Long. Pm (SUS)	60	60	60	60	60	60	60	60
Long. Pl (SUS)	0	0	0	0	1	1	1	1
Long. Q (SUS)	0	0	20	-20	-1	1	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	120.0	126.0	120.0	126.0	120.0	126.0	120.0	126.0
Pm+Pl (SUS)	121.0	127.0	121.0	127.0	120.0	126.0	120.0	126.0
Pm+Pl+Q (Total)	121.0	127.0	127.0	121.0	116.0	130.0	144.0	102.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	126.00	147.00	Passed
Pm+Pl (SUS)	127.00	220.50	Passed
Pm+Pl+Q (TOTAL)	144.01	441.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):



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Nozzle Calcs.: N2-Tube Outlet

Noz1: 12

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Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	10	3	10	3	10	3	10
Circ. Pl (SUS)	1	1	1	1	1	1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	0	0	6	-6	-2	2	8	-8
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	3.0	10.0	3.0	10.0	3.0	10.0	3.0	10.0
Pm+Pl (SUS)	5.0	12.0	4.0	11.0	5.0	12.0	4.0	11.0
Pm+Pl+Q (Total)	5.0	12.0	9.0	14.0	7.0	11.0	11.0	16.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	147.00	Passed
Pm+Pl (SUS)	12.00	220.51	Passed
Pm+Pl+Q (TOTAL)	16.00	441.01	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: Vent

Nozl: 13 9:46am Sep 9,2026

Input, Nozzle Desc: Vent

From: 30

Pressure for Reinforcement Calculations	P	5900.000	KPa.
Temperature for Internal Pressure	Temp	430	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	147.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1285.00	mm
Design Length of Section	L	999.9695	mm
Shell Finished (Minimum) Thickness	t	34.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		803.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	147.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-336 F11	
Hub Height of Integral Nozzle	h	26.1772	mm
Height of Beveled Transition	L'	23.8228	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



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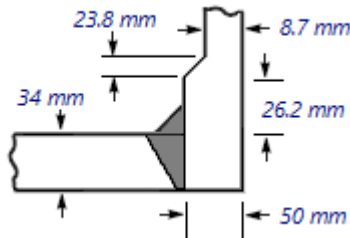
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Nozzle Calcs.: Vent

Nozl: 13

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Reinforcement CALCULATION, Description: Vent

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	42.850 mm.
Actual Thickness Used in Calculation	8.738 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5900.0 \cdot 645.5 / (147 \cdot 1.0 - 0.6 \cdot 5900.0)$
 $= 26.5487 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5900.0 \cdot 24.42 / (147 \cdot 1.0 - 0.6 \cdot 5900.0)$
 $= 1.0045 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3541 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 31.0 / 0.5773$
 $= 53.6936 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	122.3250 mm
Parallel to Vessel Wall	Rn+tn+tr	61.1625 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	64.3440 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Vent.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.0045 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm



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Nozzle Calcs.: Vent

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Wall Thickness, shell/head, internal pressure	trb1 = 29.5487 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 29.5487 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.4543 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]
 = min[6.42, max(29.5487, 4.5)]
 = 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(4.0045, 6.42)
 = 6.4200 mm

Available Nozzle Neck Thickness = $0.875 \times 8.738 = 7.645$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 7.645, tr = 0.998, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \times E^* / (tg - c) = 0.215$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 34.0, tr = 26.365, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \times E^* / (tg - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-1 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -1 °C

Weld Size Calculations, Description: Vent

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \times Wo$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 2.9178 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 286.9178 mm

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Nozzle Calcs.: N3-Shell Inlet

Nozl: 14

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Input, Nozzle Desc: N3-Shell Inlet

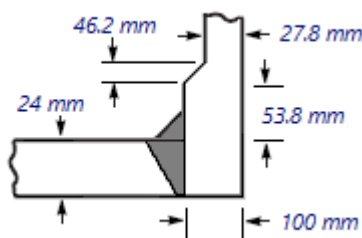
From: 40

Pressure for Reinforcement Calculations	P	4350.000	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	148.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1285.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5038.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		350.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-336 F11	
Hub Height of Integral Nozzle	h	53.8120	mm
Height of Beveled Transition	L'	46.1880	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.





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Nozzle Calcs.: N3-Shell Inlet

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Reinforcement CALCULATION, Description: N3-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	300.025 mm.
Actual Thickness Used in Calculation	27.788 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4350.0 \cdot 645.5 / (148 \cdot 1.0 - 0.6 \cdot 4350.0)$
 $= 19.3142 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4350.0 \cdot 153.01 / (148 \cdot 1.0 - 0.6 \cdot 4350.0)$
 $= 4.5784 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 72.212 / 0.5773$
 $= 125.0756 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	612.0496 mm
Parallel to Vessel Wall, opening length	d	306.0248 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	52.5000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 148.0 / 148.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 148.0 / 148.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$



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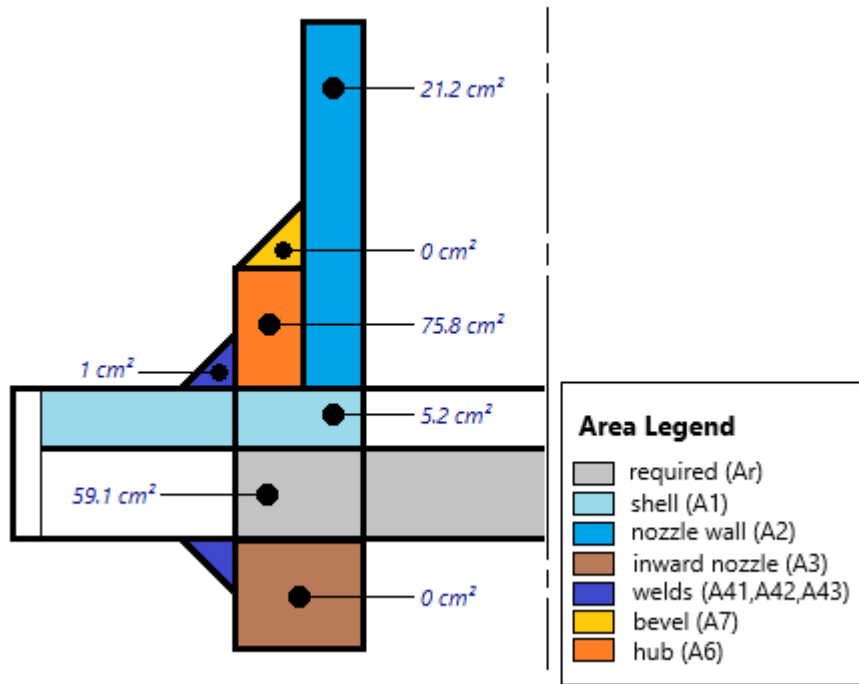
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Nozzle Calcs.: N3-Shell Inlet

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	59.107	NA	NA
Area in Shell	A1	5.159	NA	NA
Area in Nozzle Wall	A2	21.219	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	75.822	NA	NA
TOTAL AREA AVAILABLE	Atot	103.200	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (306.0248 \cdot 19.3142 \cdot 1.0 + 2 \cdot 97.0 \cdot 19.3142 \cdot 1.0 \cdot (1 - 1.)) \\
 &= 59.107 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 306.025(1.0 \cdot 21.0 - 1.0 \cdot 19.314) - 2 \cdot 97.0 \\
 &\quad (1.0 \cdot 21.0 - 1.0 \cdot 19.3142) \cdot (1 - 1.) \\
 &= 5.159 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2 \\
 &= (2 \cdot 52.5)(24.79 - 4.58) \cdot 1. \\
 &= 21.219 \text{ cm}^2
 \end{aligned}$$



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Nozzle Calcs.: N3-Shell Inlet

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(52.5, 250.0, 53.8)) \cdot (100.0 - 27.8) \cdot 1.$$

$$= 75.822 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.5784 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 22.3142 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 22.3142 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(22.3142, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(7.5784, 11.3312)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 27.788 = 24.314 \text{ mm}$ -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 23.6,	Allowable	: 148.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 346.4 N./mm ²	Passed
Occasional	: 12.4,	Allowable	: 196.8 N./mm ²	Passed
Shear	: 8.8,	Allowable	: 103.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 24.314, tr = 4.578, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.215$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 24.0, tr = 19.314, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.92$, Temp. Reduction = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-7 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -7 °C

Weld Size Calculations, Description: N3-Shell Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm



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Nozzle Calcs.: N3-Shell Inlet

Nozl: 14

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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (59.1069 - 5.1589 + 2 \cdot 97.0 \cdot 1. \cdot (1.0 \cdot 21.0 - 19.3142)) \cdot 148) \\
 &= 86351.26 \text{ Kgf}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (97.0409 + 1. - 0.0 \cdot 1.) \cdot 148 \\
 &= 147958.62 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (97.0409 + 0.0 + 1. + (40.7393)) \cdot 148 \\
 &= 209440.39 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (97.0409 + 0.0 + 1. + 0.0 + (40.7393)) \cdot 148 \\
 &= 209440.39 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 500.0248 \cdot 10.0 \cdot 0.49 \cdot 148 \\
 &= 58081. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 201.5124) \cdot (100.0 - 3.0) \cdot 0.7 \cdot 148 \\
 &= 648704. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 500.0248 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 148 \\
 &= 184202. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (58081 + 648704) = 706785 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (58081 + 0 + 184203 + 0) = 242283 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (58081 + 184203 + 0) = 242283 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 706784 Kgf, must exceed W = 86351 Kgf or W1 = 147958 Kgf



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Nozzle Calcs.: N3-Shell Inlet Nozl: 14 9:46am Sep 9, 2026

Path 2-2 = 242283 Kgf, must exceed W = 86351 Kgf or W2 = 209440 Kgf
 Path 3-3 = 242283 Kgf, must exceed W = 86351 Kgf or W3 = 209440 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4710 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 50.6385 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 324.6385 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 306.02 mm

Input Echo, WRC297 Item 1, Description: N3-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1285.000	mm	
Shell Thickness	24.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-387 11		
Vessel Cold S.I. Allowable	Smc 148.00	N./mm ²	
Vessel Hot S.I. Allowable	Smh 148.00	N./mm ²	
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	300.025	mm	
Nozzle Thickness	100.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-336 F11		
Nozzle Cold S.I. Allowable	SNmc 148.00	N./mm ²	
Nozzle Hot S.I. Allowable	SNmh 148.00	N./mm ²	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	4350.00	KPa.
Radial Load	P	-2676.93	Kgf
Circumferential Shear	Vc	1784.62	Kgf
Longitudinal Shear	Vl	1784.62	Kgf
Circumferential Moment	Mc	1748.26	Kg-m.
Longitudinal Moment	Ml	1748.26	Kg-m.
Torsional Moment	Mt	2497.50	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.000) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 500.025 + 2 \cdot 1.65 \cdot \sqrt{(656.0(24.0 - 3.0))}$$

$$= 887.350 \text{ mm}$$



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Nozzle Calcs.: N3-Shell Inlet

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WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1312.000 mm
 Nozzle Outside Diameter (d) = 500.025 mm
 Vessel Thickness used (T) = 21.000 mm
 Nozzle Thickness used (t) = 97.000 mm
 T / t = 0.216
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.012$$

Nr/P = 0.058
 Mr/P = 0.106
 M0/P = 0.032
 N0/P = 0.063
 MrD/Mc = 0.266
 NrDL/Mc = 0.092
 M0d/Mc = 0.080
 N0DL/Mc = 0.086
 MrD/Ml = 0.084
 NrDL/Ml = 0.063
 M0D/Ml = 0.024
 N0DL/Ml = 0.077

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-11	11	11	-11
Normal Pressure Stress	131	135	131	135
Outplane Stress Summary	129	133	163	123

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-39	39	39	-39
Inplane Pressure Stress	65	65	65	65
Inplane Stress Summary	63	65	150	-2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	4	4	1	1



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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	129	133	164	125
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-37	37	37	-37
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65	65	65	65

Outplane Stress Summary	37	88	124	27
-------------------------	----	----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-124	124	124	-124
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	131	135	131	135

Inplane Stress Summary	41	218	303	-15
------------------------	----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	4	4
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	42	218	304	43
------------------------	----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5



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Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	11	6	11
Hoop Stress Summary	4	9	15	21

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	1	1	7	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	4	9	16	21

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	11	6	11
Hoop Stress Summary	3	8	16	21

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-4	4	4	-4
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-1	5	11	-1



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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	5	8	16	23

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	131	135	131	135	131	135	131	135
Circ. Pl (SUS)	-2	-2	8	8	-4	-4	10	10
Circ. Q (SUS)	0	0	22	-22	-87	87	161	-161
Long. Pm (SUS)	65	65	65	65	65	65	65	65
Long. Pl (SUS)	-1	-1	7	7	-3	-3	9	9
Long. Q (SUS)	-2	2	76	-76	-26	26	48	-48
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	131.0	135.0	131.0	135.0	131.0	135.0	131.0	135.0
Pm+Pl (SUS)	129.0	133.0	139.0	143.0	127.0	131.0	141.0	145.0
Pm+Pl+Q (Total)	129.1	133.1	161.1	125.0	40.2	218.0	302.0	42.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	135.00	148.00	Passed
Pm+Pl (SUS)	145.01	222.00	Passed
Pm+Pl+Q (TOTAL)	302.05	444.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	11	6	11	6	11	6	11
Circ. Pl (SUS)	-2	-2	8	8	-3	-3	9	9



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Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	2	-2	-3	3	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	6.0	11.0	6.0	11.0	6.0	11.0	6.0	11.0
Pm+Pl (SUS)	4.0	9.0	14.0	19.0	3.0	8.0	15.0	20.0
Pm+Pl+Q (Total)	4.0	9.0	14.0	19.0	4.0	8.0	15.0	21.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	148.00	Passed
Pm+Pl (SUS)	20.00	222.00	Passed
Pm+Pl+Q (TOTAL)	21.00	443.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N4-Shell Outlet

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Input, Nozzle Desc: N4-Shell Outlet

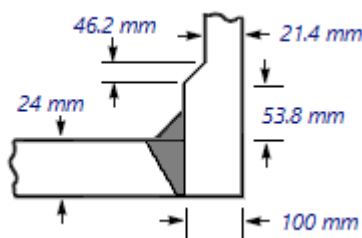
From: 40

Pressure for Reinforcement Calculations	P	4362.596	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	148.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1285.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5038.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		400.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-336 F11	
Hub Height of Integral Nozzle	h	53.8120	mm
Height of Beveled Transition	L'	46.1880	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.





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Nozzle Calcs.: N4-Shell Outlet

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Reinforcement CALCULATION, Description: N4-Shell Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 363.576 mm.
Actual Thickness Used in Calculation 21.412 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4362.6 \cdot 645.5 / (148 \cdot 1.0 - 0.6 \cdot 4362.6)$
= 19.3711 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4362.6 \cdot 184.79 / (148 \cdot 1.0 - 0.6 \cdot 4362.6)$
= 5.5455 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 78.588 / 0.5773
= 136.1181 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	739.1512	mm
Parallel to Vessel Wall, opening length	d	369.5756	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	52.5000	mm

Weld Strength Reduction Factor [fr1]:

= min(1, S_n / S_v)
= min(1, 148.0 / 148.0)
= 1.000

Weld Strength Reduction Factor [fr2]:

= min(1, S_n / S_v)
= min(1, 148.0 / 148.0)
= 1.000

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)
= min(1., 1.0)
= 1.000



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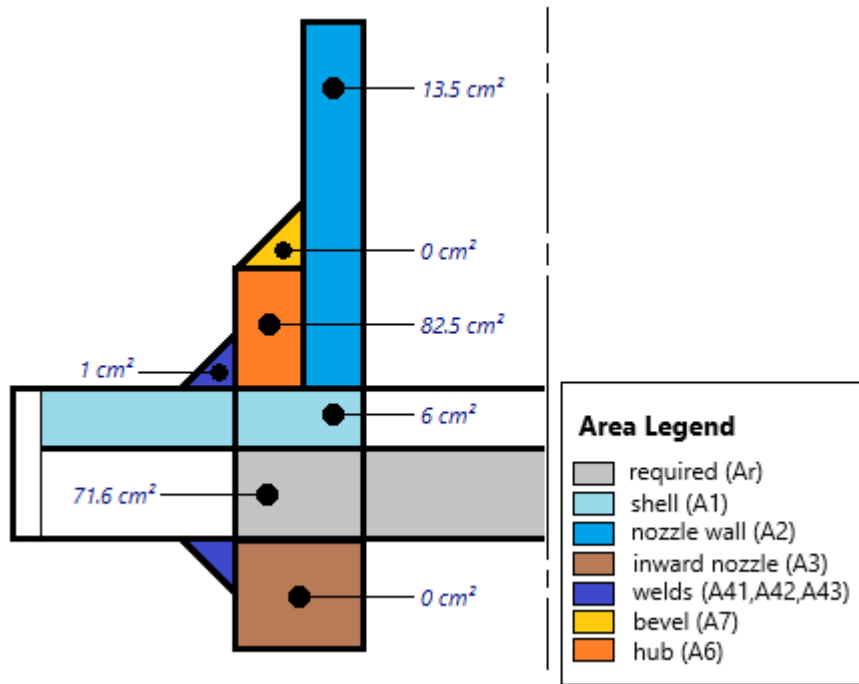
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Nozzle Calcs.: N4-Shell Outlet

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	71.592	NA	NA
Area in Shell	A1	6.020	NA	NA
Area in Nozzle Wall	A2	13.510	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	82.516	NA	NA
TOTAL AREA AVAILABLE	Atot	103.045	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (369.5756 \cdot 19.3711 \cdot 1.0 + 2 \cdot 97.0 \cdot 19.3711 \cdot 1.0 \cdot (1 - 1.)) \\
 &= 71.592 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 369.576(1.0 \cdot 21.0 - 1.0 \cdot 19.371) - 2 \cdot 97.0 \\
 &\quad (1.0 \cdot 21.0 - 1.0 \cdot 19.371) \cdot (1 - 1.) \\
 &= 6.020 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tnp)(tn - trn)fr2 \\
 &= (2 \cdot 52.5)(18.41 - 5.55) \cdot 1. \\
 &= 13.510 \text{ cm}^2
 \end{aligned}$$



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Nozzle Calcs.: N4-Shell Outlet

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(52.5, 250.0, 53.8)) \cdot (100.0 - 21.4) \cdot 1.$$

$$= 82.516 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 8.5455 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 22.3711 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 22.3711 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.331, \max(22.3711, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(8.5455, 11.3312)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 21.412 = 18.736 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 34.7,	Allowable	: 148.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 335.3 N./mm ²	Passed
Occasional	: 20.9,	Allowable	: 196.8 N./mm ²	Passed
Shear	: 10.7,	Allowable	: 103.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 18.736, tr = 5.545, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.352$, Temp. Reduction = 76 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 24.0, tr = 19.371, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.922$, Temp. Reduction = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-6 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -6 °C

Weld Size Calculations, Description: N4-Shell Outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm



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Nozzle Calcs.: N4-Shell Outlet

Nozl: 15

9:46am

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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (71.5917 - 6.0199 + 2 \cdot 97.0 \cdot 1. \cdot (1.0 \cdot 21.0 - 19.3711)) \cdot 148) \\
 &= 103726.68 \text{ Kgf}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (96.0255 + 1. - 0.0 \cdot 1.) \cdot 148 \\
 &= 146426.23 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (96.0255 + 0.0 + 1. + (40.7393)) \cdot 148 \\
 &= 207907.98 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (96.0255 + 0.0 + 1. + 0.0 + (40.7393)) \cdot 148 \\
 &= 207907.98 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 563.5756 \cdot 10.0 \cdot 0.49 \cdot 148 \\
 &= 65463. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 233.2878) \cdot (100.0 - 3.0) \cdot 0.7 \cdot 148 \\
 &= 750995. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 563.5756 \cdot (24.0 - 3.0) \cdot 0.74 \cdot 148 \\
 &= 207614. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (65463 + 750995) = 816457 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (65463 + 0 + 207614 + 0) = 273076 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (65463 + 207614 + 0) = 273076 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 816457 Kgf, must exceed W = 103726 Kgf or W1 = 146426 Kgf



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Nozzle Calcs.: N4-Shell Outlet Nozl: 15 9:46am Sep 9, 2026

Path 2-2 = 273076 Kgf, must exceed W = 103726 Kgf or W2 = 207907 Kgf
 Path 3-3 = 273076 Kgf, must exceed W = 103726 Kgf or W3 = 207907 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4710 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 65.0904 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 339.0903 mm

Input Echo, WRC297 Item 1, Description: N4-Shell Outle:

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1285.000	mm	
Shell Thickness	24.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-387 11		
Vessel Cold S.I. Allowable	Smc 148.00	N./mm ²	
Vessel Hot S.I. Allowable	Smh 148.00	N./mm ²	
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	363.576	mm	
Nozzle Thickness	100.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-336 F11		
Nozzle Cold S.I. Allowable	SNmc 148.00	N./mm ²	
Nozzle Hot S.I. Allowable	SNmh 148.00	N./mm ²	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	4362.60	KPa.
Radial Load	P	-3059.35	Kgf
Circumferential Shear	Vc	2039.57	Kgf
Longitudinal Shear	Vl	2039.57	Kgf
Circumferential Moment	Mc	2283.43	Kg-m.
Longitudinal Moment	Ml	2283.43	Kg-m.
Torsional Moment	Mt	3262.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.636) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 563.576 + 2 \cdot 1.65 \cdot \sqrt{656.0(24.0 - 3.0)}$$

$$= 950.901 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1312.000 mm
Nozzle Outside Diameter	(d) =	563.576 mm



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Nozzle Calcs.: N4-Shell Outlet Nozl: 15 9:46am Sep 9,2026

Vessel Thickness used (T) = 21.000 mm
 Nozzle Thickness used (t) = 97.000 mm
 T / t = 0.216
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.395$$

Nr/P = 0.054
 Mr/P = 0.095
 M0/P = 0.029
 N0/P = 0.060
 MrD/Mc = 0.261
 NrDL/Mc = 0.091
 M0d/Mc = 0.078
 N0DL/Mc = 0.086
 MrD/Ml = 0.076
 NrDL/Ml = 0.056
 M0D/Ml = 0.022
 N0DL/Ml = 0.068

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-12	12	12	-12
Normal Pressure Stress	131	136	131	136
Outplane Stress Summary	129	134	166	122

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	38	-38	38	-38
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-41	41	41	-41
Inplane Pressure Stress	65	65	65	65
Inplane Stress Summary	62	67	154	-5

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	1	1

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------



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Nozzle Calcs.: N4-Shell Outlet

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Two * Max Shear Stress	129	134	166	128
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-42	42	42	-42
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	65	65	65	65
Outplane Stress Summary	31	92	131	23

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	38	-38	38	-38
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-141	141	141	-141
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	131	136	131	136
Inplane Stress Summary	25	234	323	-31

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	1	1	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	32	234	323	55

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-6	-6	6	6
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	13	7	13
Hoop Stress Summary	5	11	18	23



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Nozzle Calcs.: N4-Shell Outlet

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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	2	2	8	3

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	6	11	18	23

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-7	-7	7	7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	13	7	13
Hoop Stress Summary	4	9	19	24

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-1	6	12	-1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside



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Nozzle Calcs.: N4-Shell Outlet

Nozl: 15 9:46am

Sep 9, 2026

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	5	9	19	25

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	131	136	131	136	131	136	131	136
Circ. Pl (SUS)	-2	-2	10	10	-5	-5	11	11
Circ. Q (SUS)	-1	1	23	-23	-103	103	179	-179
Long. Pm (SUS)	65	65	65	65	65	65	65	65
Long. Pl (SUS)	-2	-2	8	8	-3	-3	11	11
Long. Q (SUS)	-3	3	79	-79	-31	31	53	-53
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	131.0	136.0	131.0	136.0	131.0	136.0	131.0	136.0
Pm+Pl (SUS)	129.0	134.0	141.0	146.0	126.0	131.0	142.0	147.0
Pm+Pl+Q (Total)	128.2	135.2	164.3	129.1	31.5	234.0	321.1	55.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	136.00	148.00	Passed
Pm+Pl (SUS)	147.01	222.00	Passed
Pm+Pl+Q (TOTAL)	321.08	444.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	13	7	13	7	13	7	13
Circ. Pl (SUS)	-2	-2	10	10	-3	-3	11	11
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1



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Nozzle Calcs.: N4-Shell Outlet

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Long. Q (SUS)	0	0	2	-2	-4	4	6	-6
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	7.0	13.0	7.0	13.0	7.0	13.0	7.0	13.0
Pm+Pl (SUS)	5.0	11.0	17.0	23.0	4.0	10.0	18.0	24.0
Pm+Pl+Q (Total)	5.0	11.0	17.0	23.0	6.0	10.0	18.0	26.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	13.00	148.00	Passed
Pm+Pl (SUS)	24.00	222.00	Passed
Pm+Pl+Q (TOTAL)	26.00	443.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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ASME TS Calc:

Case: 1 9:46a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: U-Tube

Shell Data:

Main Shell Description: Main Shell-1

Shell Maximum Design Pressure	Psd,max	4350.00 KPa.
Shell Maximum Operating Pressure	Psox,max	4350.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	24.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1285.000 mm
Shell Temperature for Internal Pressure	Ts	275.00 °C
Shell Material		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	148.00 N./mm ²
Shell Allowable Stress at Ambient		148.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	5900.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	5900.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	34.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1285.000 mm
Channel Design Temperature	TEMPC	430.00 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	147.00 N./mm ²
Channel Allowable Stress at Ambient		148.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	626
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	6000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5815.00 mm
Design Temperature of the Tubes		430.00 °C
Tube Material		SA-213 T11
Tube Material UNS Number		K11597
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		98.98 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	148.20 N./mm ²
Tube Pitch (Center to Center Spacing)	P	29.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	618.975 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm



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ASME TS Calc:

Case: 1 9:46a Sep 9,2026

Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	4763.3 cm ²
Perimeter of Tube Layout	Cp	3248.0 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	430.00 °C
Tubesheet Material		SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

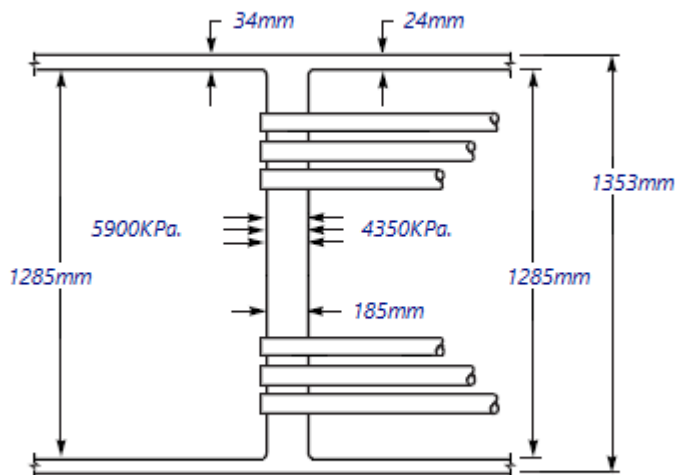
Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	147.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	148.00 N./mm ²
Thickness of Tubesheet	h	185.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1353.000 mm

Area of the Untubed Lanes	AL	457.2 cm ²
---------------------------	----	-----------------------

Junction Stress Reduction option	Analyze as Simply Supported
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes
Channel Side Min. Design Pressure	Ptd,min 103.0000 KPa.

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group C



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ASME TS Calc:

Case: 1 9:46a Sep 9,2026

Elastic Mod. at Design Temperature	275.0 °C	0.18798E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	430.0 °C	0.17617E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TM-1 Material Group C

Elastic Mod. at Tubsht. Design Temp.	430.0 °C	0.17617E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	430.0 °C	0.17617E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(5900.0 \cdot (19.05/2 - 0)) / (98.98 \cdot 1.0 + 0.4 \cdot 5900.0)$

= $0.5546 + 0.0000 = 0.5546$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	CS-2	at	430.00 °C
Elastic Modulus for Material			156378400.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	6433.98	9.03	50.0000	0.0134948	77.47

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 12139.8154$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.0222	19.05	6433.98	18.64	50.0000	0.0031674	62.25

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 62.2479) / (3 \cdot 18.6356) = 4453.4248$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.0222 mm
Allowable Internal Pressure at Corroded thickness	24057.11 KPa.
Required Internal Design Pressure	5900.00 KPa.
Allowable External Pressure at Corroded thickness	12139.82 KPa.
Required External Design Pressure	4453.00 KPa.
Required Thickness due to Shell Side pressure	1.0222 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

Note:

The Tubesheet stress reduction option was Simply Supported. Performing the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:



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ASME TS Calc:

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Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned}
 &= 2 \cdot r_o + d_t \\
 &= 2 \cdot 618.975 + 19.05 \\
 &= 1257.000 \text{ mm}
 \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned}
 &= (p - d_t) / p \\
 &= (29.0 - 19.05) / 29.0 \\
 &= 0.343
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned}
 &= D_s / D_o \text{ (Configurations a, b, c)} \\
 &= 1285.0 / 1257.0 \\
 &= 1.0223
 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned}
 &= D_c / D_o \text{ (Configurations a, e, f)} \\
 &= 1285.0 / 1257.0 \\
 &= 1.0223
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned}
 &= l_{tx} / h \\
 &= 0 / 185.0 \\
 &= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}
 \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned}
 &= \max(d_t - 2t_t(E_t T / E) \cdot (S_t T / S) \cdot (\rho); d_t - 2t_t) \\
 &= \max(19.05 - 2 \cdot 2.11 (0.17617E+09 / 1.7617E+09) \cdot \\
 &\quad (98 / 147) \cdot (0); 19.05 - 2 \cdot 2.11) \\
 &= 19.0500 \text{ mm}
 \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned}
 &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot \rho) / (\pi \cdot D_o^2)) } \\
 &= 29.0 / \sqrt{ (1 - 4 \cdot \min(457.2; 4 \cdot 1257.0 \cdot 29.0) / (\pi \cdot 1257.0^2)) } \\
 &= 29.5494 \text{ mm}
 \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned}
 &= (p^* - d^*) / p^* \\
 &= (29.5494 - 19.05) / 29.5494 \\
 &= 0.3553
 \end{aligned}$$

$h/p = 6.379310$; $\mu^* = 0.355318$; $E^* = 62405232 \text{ KPa}$.

$E^*/E = 0.354224$; $v^* = 0.324368$

Note: As h/p (6.379) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$\begin{aligned}
 &= ((12(1 - v_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) } \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((1285.0 + 24.0) 24.0) } \\
 &= 0.0103 \text{ 1/mm}
 \end{aligned}$$

Shell coefficient [k_s]:

$$\begin{aligned}
 &= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2)) \\
 &= 0.01 \cdot 0.18798E+09 \cdot 24.0^3 / (6(1 - 0.3^2)) \\
 &= 4881341952.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$



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ASME TS Calc:

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$$= 6(1285.0 + 24.0) \cdot 0.48813E+1 / 185.0^3 \cdot (1 + 185.0 \cdot 0.01 + (185.0 \cdot 0.01)^2 / 2)$$

$$= 28442658.0 \text{ KPa.}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 1285.0^2 / (4 \cdot 0.18798E+09 \cdot 24.0) \cdot (1 - 1285.0 \cdot 0.3 / (2(1285.0 + 24.0)))$$

$$= 0.78021E-01 \text{ mm/N./mm}^2$$

Parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.0223 \cdot 0.48813E+1 \cdot 0.0103 \cdot 0.078021 (1 + 185.0 \cdot 0.0103)$$

$$= 11569.9141 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4} / \sqrt{(D_c + t_c)t_c})$$

$$= ((12(1 - 0.3^2))^{1/4} / ((1285.0 + 34.0)34.0))$$

$$= 0.0086 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.009 \cdot 0.17617E+09 \cdot 34.0^3 / (6(1 - 0.3^2))$$

$$= 10886309888.0000 \text{ KPa.} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1285.0 + 34.0 / \cos(0)) \cdot 0.10886E+11 / 185.0^3 \cdot (1.0 + 185.0 \cdot 0.009 \cdot (1.0 + 1.0) / 2 + (185.0 \cdot 0.009)^2 / 2 \cdot 1.0)$$

$$= 52373424.0 \text{ KPa.}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1285.0^2 / (4 \cdot 0.17617E+09 \cdot 34.0) \cdot (1 - 1285.0 \cdot 0.3 / (2(1285.0 + 34.0)))$$

$$= 0.58842E-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.022 \cdot 0.10886E+11 [0.058842 (1.0 + 0.009 \cdot 1.0 \cdot 185.0) - 0 (1.0 / 185.0 + 1.0 \cdot 0.009 / 2)]$$

$$= 14548.9092 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1353.0 / 1257.0$$

$$= 1.0764$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.32) / 62405232 \cdot (28442658 + 52373424 + 0.17617E+09 \cdot \ln(1.08))$$

$$= 1.0153$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$



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ASME TS Calc:

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$$= 1257.0^2/16 \cdot [(1.022 - 1) \cdot (1.022^2 + 1) \cdot 4350.0 - (1.022 - 1) \cdot (1.022^2 + 1) \cdot 5900.0]$$

$$= -6972836.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -6972836.0 + 14548.909 \cdot 5900.0 - 11569.914 \cdot 4350.0$$

$$= 28536606.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^* - D_o^2/32 \cdot F(P_s - P_t)) / (1 + F)$$

$$= (28536606 - 1257.0^2/32 \cdot 1.015(4350.0 - 5900.0)) / (1 + 1.02)$$

$$= 52717688.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= 52717688 + 1257.0^2/64 \cdot (3 + 0.324) \cdot (4350.0 - 5900.0)$$

$$= -74495240.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|52717688|; |-0.7E+08|)$$

$$= 74495240.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 74495240 / (0.3553(185.0 - 8.0)^2)$$

$$= 40.1552 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 147.0$$

$$= 294.01 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 69126312 / (0.3553(71.0484 - 8.0)^2)$$

$$= 293.6669 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 71.0484 + 0 + 0$$

$$= 71.0484 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(71.0484; 5.6338)$$

$$= 71.0484 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [|Ps - Pt|]:

$$= |P_s - P_t|$$

$$= |4350.0 - 5900.0|$$

$$= 1550.000 \text{ KPa}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.343 \cdot 185.0 / 1257.0 \cdot \min(0.8 \cdot 147.005; 0.533 \cdot 223.003)$$

$$= 23.754 \text{ N./mm}^2$$



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| Ps - Pt | is ≤ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot | Ps - Pt | + Cats + Catc$$

$$= 1 / (4 \cdot 0.343 \cdot 117.604) \cdot (4 \cdot 4763.3 / 3248.0) \cdot | 4350.0 - 5900.0 | + 0$$

$$= 5.6338 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$= Ds^2 / (4 \cdot ts (Ds + ts)) \cdot Ps$$

$$= 1285.0^2 / (4 \cdot 24.0 (1285.0 + 24.0)) \cdot 4350.0$$

$$= 57.1623 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot ks / ts^2 \cdot [\beta S \cdot P's + 6(1 - \nu^*) / E^* \cdot Do / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 1097394 / 24.0^2 \cdot [0.01 \cdot 0.3383 + 6(1 - 0.324) / 62405232 \cdot 1257.0 / 185.0^3 \cdot (1 + 185.0 \cdot 0.01 / 2) \cdot (52717688 + 1257.0^2 / 32 \cdot (4350.0 - 5900.0))]$$

$$= 146.577 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= | \sigma_s, m | + | \sigma_s, b | \text{ (Should be } \leq \text{ SPS, s)}$$

$$= 203.740 \text{ N./mm}^2 \text{ (Should be } \leq 498.02)$$

Axial Channel Membrane Stress [σ_c, m]:

$$= Dc^2 / (4 \cdot tc (Dc + tc) \cdot \cos(\theta)) \cdot Pt$$

$$= 1285.0^2 / (4 \cdot 34.0 (1285.0 + 34.0) \cdot \cos(0)) \cdot 5900.0$$

$$= 54.312 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot kc / tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\Theta) \cdot Pt - 6((1 - \nu^*) / E^*) \cdot Do / h^3 \cdot (Cm\Theta + h \cdot \beta_c / 2 \cdot Cm\Delta) \cdot (Mp + Do^2 / 32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 2447395 / 34.0^2 \cdot [(0.058842 \cdot 1.0 \cdot 0.009 - 0 \cdot 1.0) \cdot 5900.0 - 6((1 - 0.324) / 62405232) \cdot 1257.0 / 185.0^3 \cdot (1.0 + 0.009 \cdot 185.0 / 2 \cdot 1.0) \cdot (52717688 + 1257.0^2 / 32 \cdot (4350.0 - 5900.0))]$$

$$= 199.542 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_c, m | + | \sigma_c, b | \text{ (Should be } \leq \text{ SPS, c)}$$

$$= 253.855 \text{ N./mm}^2 \text{ (Should be } \leq 446.03)$$

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:

$$\beta_s, ks, \lambda_s, \delta_s \text{ are all } = 0$$

Intermediate Channel side parameters for the Simply Supported condition:

$$\beta_c, kc, \lambda_c, \delta_c \text{ are all } = 0$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:



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ASME TS Calc:

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Coefficient [F]:

$$\begin{aligned}
 &= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\
 &= (1 - 0.32) / 62405232 \cdot (0 + 0 + 0.17617E+09 \cdot \ln(1.08)) \\
 &= 0.1404
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned}
 &= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt] \\
 &= 1257.0^2/16 \cdot [(1.022 - 1) \cdot (1.022^2 + 1) \cdot 4350.0 - (1.022 - 1) \cdot (1.022^2 + 1) \cdot 5900.0] \\
 &= -6972836.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned}
 &= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps \\
 &= -6972836.0 + 0 \cdot 5900.0 - 0 \cdot 4350.0 \\
 &= -6972836.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned}
 &= Mp + Do^2/64 \cdot (3 + \nu^*) \cdot (Ps - Pt) \\
 &= 3306416 + 1257.0^2/64 \cdot (3 + 0.324) \cdot (4350.0 - 5900.0) \\
 &= -123906512.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= |Mo| \\
 &= |-0.1E+09| \\
 &= 123906512.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu^* (h - hg')^2) \\
 &= 6 \cdot 0.12391E+09 / (0.3553 (185.0 - 8.0)^2) \\
 &= 66.7893 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 \cdot S \\
 &= 2 \cdot 147.0 \\
 &= 294.01 \text{ N./mm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot M / (\mu^* (h - hg')^2) \\
 &= 6 \cdot 0.12391E+09 / (0.3553 (92.3856 - 8.0)^2) \\
 &= 293.8443 \text{ N./mm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + Cats + Catc \\
 &= 92.3856 + 0 + 0 \\
 &= 92.3856 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(HreqB; HreqS) \\
 &= \max(92.3856; 12.0722) \\
 &= 92.3856 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$\begin{aligned}
 &= |Ps - Pt| \\
 &= |4350.0 - 5900.0| \\
 &= 1550.000 \text{ KPa}
 \end{aligned}$$



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ASME TS Calc:

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Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h/Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)$$

$$= 4 \cdot 0.343 \cdot 185.0/1257.0 \cdot \min(0.8 \cdot 147.005; 0.533 \cdot 223.003)$$

$$= 23.754 \text{ N./mm}^2$$

$|Ps - Pt|$ is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1/(4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap/Cp) \cdot |Ps - Pt| + Cats + Catc$$

$$= 1/(4 \cdot 0.343 \cdot 117.604) \cdot (4 \cdot 4763.3/3248.0) \cdot |4350.0 - 5900.0| + 0$$

$$= 5.6338 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$= Ds^2 / (4 \cdot ts (Ds + ts)) \cdot Ps$$

$$= 1285.0^2 / (4 \cdot 24.0 (1285.0 + 24.0)) \cdot 4350.0$$

$$= 57.1623 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_s, b]:

$$= 6 \cdot ks/ts^2 \cdot [\beta S \cdot P's + 6(1 - \nu^*)/E^* \cdot Do/h^3 \cdot (1 + h \cdot \beta S/2) \cdot (Mp + Do^2/32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 0/24.0^2 \cdot [0 \cdot 0.3383 + 6(1 - 0.324)/62405232 \cdot 1257.0/185.0^3 \cdot (1 + 185.0 \cdot 0/2) \cdot (3306416 + 1257.0^2/32 \cdot (4350.0 - 5900.0))]$$

$$= -0.000 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_s, m| + |\sigma_s, b| \text{ (Should be } \leq \text{SPS, s)}$$

$$= 57.162 \text{ N./mm}^2 \text{ (Should be } \leq 498.02)$$

Axial Channel Membrane Stress [σ_c, m]:

$$= Dc^2 / (4 \cdot tc (Dc + tc)) \cdot \cos(\theta) \cdot Pt$$

$$= 1285.0^2 / (4 \cdot 34.0 (1285.0 + 34.0)) \cdot \cos(0) \cdot 5900.0$$

$$= 54.312 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_c, b]:

$$= 6 \cdot kc/tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\Theta) \cdot Pt - 6((1 - \nu^*)/E^*) \cdot Do/h^3 \cdot (Cm\Theta + h \cdot \beta_c/2 \cdot Cm\Delta) \cdot (Mp + Do^2/32 \cdot (Ps - Pt))]$$

$$= 6 \cdot 0/34.0^2 \cdot [(0 \cdot 1.0 \cdot 0 - 0 \cdot 1.0) \cdot 5900.0 - 6((1 - 0.324)/62405232) \cdot 1257.0/185.0^3 \cdot (1.0 + 0 \cdot 185.0/2 \cdot 1.0) \cdot (3306416 + 1257.0^2/32 \cdot (4350.0 - 5900.0))]$$

$$= 0.000 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_c, m| + |\sigma_c, b| \text{ (Should be } \leq \text{SPS, c)}$$

$$= 54.312 \text{ N./mm}^2 \text{ (Should be } \leq 446.03)$$

Analysis Finished.

Tube Metal Area [A_t]:

$$= \pi (do - t) t$$

$$= \pi (19.05 - 2.11) 2.11$$

$$= 1.123 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:



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ASME TS Calc:

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$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (19.05 - 2.11) 98.982$$

$$= 1133.373 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 98.98$$

$$= 1133.3734 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 98.982 / 98.982$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 98.98$$

$$= 753.0200 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) 1.0 \cdot 1.0} - 0.75 \cdot 19.05 \right)$$

$$= 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm
Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 1133.37 is >= tube strength 1133.37 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	66.8 <=	294.0 N./mm ²	Ok
Tubesheet shear stress	3.6 <=	117.6 N./mm ²	Ok
Stress in Shell at Tubesheet	57.2 <=	498.0 N./mm ²	Ok
Stress in Channel at Tubesheet	54.3 <=	446.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	92.386	185.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 8 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	172.597	...	254 294 14 118	Ps+Pt D1	Ok
D2uc	150.998	...	192 294 10 118	Ps+Pt D2	Ok
D3uc	92.386	...	67 294 4 118	Ps+Pt D3	Ok
D4uc	29.738	...	4 294 ... 118	Ps+Pt D4	Ok
D1c	176.484	...	266 294 14 118	Ps+Pt-c D1	Ok



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ASME TS Calc:

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D2c	154.767	...	201	294	11	118	Ps+Pt-c D2	Ok
D3c	95.825	...	70	294	4	118	Ps+Pt-c D3	Ok
D4c	32.793	...	5	294	...	118	Ps+Pt-c D4	Ok

Max: 176.4838 ... mm 0.904 0.120 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.			Shell	Stress	Channel	Stress	Pass
Case#	Shell	Channel	Stress	Allwd	Stress	Allwd	Fail
D1uc	...	...	...	498	54	446	Ok
D2uc	...	...	57	498	1	446	Ok
D3uc	...	...	57	498	54	446	Ok
D4uc	...	...	...	498	1	446	Ok
D1c	...	...	...	498	60	446	Ok
D2c	...	...	66	498	1	446	Ok
D3c	...	...	66	498	60	446	Ok
D4c	...	...	...	498	1	446	Ok
Max	...	...		0.132		0.134	

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	176.484	185.000	Ok
Tube thickness :	1.022	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Given Shell Thickness: 24.0000 mm

Given Channel Thickness: 34.0000 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	6522.96	1.000	6522.96	1.000
Tubesheet Shear Stress	49247.62	1.000	49247.62	1.000
Shell Stress (Axial, Junction)	17063.60	1.000	5338.95	1.000
Tubesheet-Channel Junction	6208.74	1.000	13772.87	1.000
Tube Pressure Stress	24057.10	1.000	12139.80	1.000
Minimum MAWP	6208.74		5338.95	



Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	6869.23	1.000	6869.23	1.000
Tubesheet Shear Stress	51242.13	1.000	51242.13	1.000
Shell Stress (Axial, Junction)	26274.80	1.000	7826.47	1.000
Tubesheet-Channel Junction	9762.50	1.000	22374.79	1.000
Tube Pressure Stress	28678.83	1.000	19233.21	1.000
Minimum MAPnc	6869.23		6869.23	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\
 &= \max(185.0/4, \min(185.0, 24.0)) \\
 &= 46.250 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned}
 &= \min(\text{ps}/\text{Shellside MAPnc}, 1) \\
 &= \min(4350.0/6869.23, 1) \\
 &= 0.633
 \end{aligned}$$

Thickness Ratio = 0.633, Temperature Reduction per Fig. UCS 66.1 = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	15 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-5 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$\begin{aligned}
 &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness})) \\
 &= \max(185.0/4, \min(185.0, 34.0)) \\
 &= 46.250 \text{ mm}
 \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned}
 &= \min(\text{pt}/\text{Tubeside MAPnc}, 1) \\
 &= \min(5900.0/6869.23, 1) \\
 &= 0.859
 \end{aligned}$$

Thickness Ratio = 0.859, Temperature Reduction per Fig. UCS 66.1 = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	15 °C
Min Metal Temp. at Required thickness (UCS 66.1)	7 °C

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: